

BWRX-300 REACTOR BUILDING

DIAPHRAGM PLATE STEEL-PLATE COMPOSITE

GENERAL NOTES & STANDARD DETAILS

SHEET	SHEET NAME	REV. DATE	REV.
G100	COVER SHEET - GENERAL NOTES & DETAILS	7/15/2026	11
G101	GENERAL NOTES	7/15/2026	11
G102	GENERAL NOTES	04/06/2026	10
G103	GENERAL WELDING AND STUD TABLES	7/15/2026	11
G104	STANDARD DETAILS	7/15/2026	11
G104.1	STANDARD DETAILS	7/15/2026	11
G105	STANDARD DETAILS	7/15/2026	11
G106	BELOW GRADE FLOOR TO SC WALL CONNECTION DETAILS	7/15/2026	11
G107	BELOW GRADE FLOOR TO SHAFT WALL CONNECTION DETAILS (TYPES 1 & 2)	7/15/2026	11
G108	BELOW GRADE FLOOR TO SHAFT WALL CONNECTION DETAILS (TYPE-3)	7/15/2026	11
G109	STANDARD DETAILS	7/15/2026	11
G110	STANDARD DETAILS	7/15/2026	11
G111	STANDARD DETAILS	04/06/2026	10

COMPREHENSIVE PACKAGE LIST	
PLM DOCUMENT NUMBER	PACKAGE TITLE
008N9536	BWRX-300 REACTOR BUILDING DIAPHRAGM PLATE STEEL-PLATE COMPOSITE GENERAL NOTES AND DETAILS
009N0961	BWRX-300 REACTOR BUILDING DIAPHRAGM PLATE STEEL-PLATE COMPOSITE BASEMAT DRAWING SET
009N0962	BWRX-300 REACTOR BUILDING DIAPHRAGM PLATE STEEL-PLATE COMPOSITE LEVEL 1 DRAWING SET

HOLD LIST - GENERAL NOTES
1. "DEFERRED VERIFICATION" STATUS IS DUE TO OPEN ITEMS IN SUPPORTING DOCUMENTS LISTED IN TABLE 1-1.
ANALYSIS AND DESIGN OPEN ITEMS ASSOCIATED WITH SUPPORTING DOCUMENTS LISTED IN TABLE 1-1 OF 008N9536 SHEET G101

REVISION SUMMARY:	
1.	ADMINISTRATIVE REVISION SEE CA-00050271
2.	ADMINISTRATIVE REVISION SEE CA-00051178 - G101 - ADDED "FABRICATION" TO GENERAL NOTE 1.3.1 - G101 - MOVED NOTES 1.11.1, 1.11.2 TO 1.12.1, 1.12.2 - G101 - REVISIONS TO 1.13 TOLERANCES
3.	DESIGN REVISION SEE CA-00060353 - G103 - REMOVE CENTER MODULE HOLDS - G101 - REVISED 1.3.5, 1.3.6 DELETED 1.3.7, 1.3.8 REMOVED HOLD.
4.	DESIGN REVISION SEE CA-00053020 - G101 - REVISE NOTE 1.3.6 - G103 - REVISED TABLE 1-5 AND DETAILS REVISED TABLE 1-6 AND NOTES FOR TABLE 1-8 - G104 - REMOVED REBAR SPLICE DETAILS ADDED FLOOR AND WALL SPLICE DETAILS - ADDED DETAILS FOR WING WALL TO RB OUTERWALL AND SCCV WALL CONNECTIONS - G105 - ADDED DETAILS FOR STAIRWALL AND ELEVATOR WALL TO RB EXTERIOR WALL CONNECTION
5.	DESIGN REVISION SEE CA-00054674 - G101 - ADD REQUIREMENT 1.14.3 - G104 - REVISE NEW ANNOTATIONS ON DETAILS D AND E - G105 - REVISE NEW ANNOTATIONS ON DETAILS A, B, AND C - G100-G105 - UPDATED TITLE BLOCK
6.	DESIGN REVISION SEE CA-00055449 - G101 - ADD NOTE 1.13.3 - G102 - UPDATED REVISION SYMBOL TO BE CONSISTENT WITH GEH DRAFTING MANUAL
7.	DESIGN REVISION SEE CA-00058765 - G100-106 UPDATED TITLE BLOCK - G103 - UPDATES TO TABLES FOR ADDITION OF NEXT LEVELS - G104 - UPDATED STUD CALL OUT IN DETAIL D AND E. - G105 - UPDATED STUD CALL OUT IN DETAIL A, B AND C. REVISED NOTE 2 AND ADDED NOTE 3.
8.	DESIGN REVISION SEE CA-00060353 - G103 - TABLE 1-5 WELD INSPECTION REQUIREMENTS DATA DELETED - G103 - TABLE 1-7 TYPO CORRECTION - G103 - TABLE 1-8 NEW DATA ADDED - G104 - DETAIL "D" & "E" UPDATED - G104 - NEW DETAIL H ADDED - G104 - PLACEHOLDER DETAIL "I" ADDED - G106-G108 - NEW SHEETS ADDED
9.	DESIGN REVISION SEE CA-00061148 - G103 - REVISION TO RESOLVE ISS-0014764 - G104.1 - NEW SHEET ADDED - G105 - DETAIL "A" REVISION - G106 - BUILT UP BEAM MODIFIED, TYPO CORRECTIONS, AND NEW NOTE 11 - G107 - NEW SEAT BRACKET TYPE-2A DETAIL - G108 - FULLY REVISED - G109-G111 - NEW SHEETS ADDED

10.	DESIGN REVISION SEE CA-00063942 - G101 - TYPO, TABLE MODIFICATION AND ADDING NEW CONTENT FOR CONCRETE. - G102 - MODULE TAGGING BREAKDOWN TABLE UPDATED - G103 - TABLES 1-5, 1-6 & 1-7 NOTES MINOR UPDATES. TABLE 1-7 NEW NOTES 6, 7 & 8 ADDED. DETAIL 3 NEW NOTES ADDED. - G104 - DETAILS D & E ANNOTATIONS COORECTIONS. - G104.1 - GENERAL ANNOTATIONS CORRECTIONS & DIMENSIONS FOR SEVERAL DETAILS. - G105 - DETAIL "A" REVISION WITH NEW NOTES OF DOUBLER PLATE CONNECTION REQUIREMENTS. NEW CONNECTION NOTES 5 - 7. - G106 - UPDATES ON SEVERAL CONNECTION DIMENSIONS. NEW NOTE 11 & 12. - G108 - DETAIL F REVISED. - G109 - TABLE 1-11 UPDATED WITH NEW COLUMNS AND NOTES REVISED. DETAILS E, F, H REVISED. NEW DETAIL J & K. - G110 - DETAILS A THRU E REVISED. TABLE 1-13 & NOTE 3 UPDATED. NOTES 7, 8 & 9 ADDED. - G111 - DETAIL G THRU N MINOR UPDATES. NOTE ADDED ON DETAIL G and H. DETAIL L NEW TITLE NAME. DETAIL N NEW DIMENSION UPDATE.
11.	DESIGN REVISION SEE CA-00066870 - G101 - TABLE 1-4 AND NOTE 1.14 TITLE UPDATED. NEW NOTES 1.14.4 AND 14.14.5 ADDED. - G103 - TABLES 1-6, 1-7 AND 1-8 UPDATED. DETAIL 2 SEVERAL FEATURES REVISED. - G104.1 - DETAIL A REVISED AND NEW NOTES 3 & 4 ADDED. - G105 - DETAIL A REPLACED @ WITH & IN WELDED STUD CALLOUT. REFERENCE PARENTHESIS ADDED IN DETAIL A DIMENSIONS FOR DOUBLER PLATE. - G106 - DETAIL B ANNOTATION ADDED, AND GRIDS ADDED IN KEYPLAN. - G107 - DETAIL DRAWINGS A, B, C, D & F REVISED. TYPE 2-A REMOVED -- DETAILS H, I & J DELETED. NEW NOTES 8 THRU 15. - G108 - DETAIL D WITH FEW DIMENSIONS REVISED. NEW NOTE 6 ADDED. - G109 - DETAIL J REVISED DIMENSION FROM 206 TO 203, DETAIL K UPDATED DETAIL TITLE. TABLE 1-12 REVISED WITH NEW NOTES AND FEW DIMENSIONS UPDATED. NEW TABLE 1-13 ADDED. - G110 - DETAIL B UPDATED WALL ANGLE DIMENSION AND REMOVED WING AND REPLACED WITH STAIR. TABLE 1-13 CHANGED TO 1-14.

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U71

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07/15/2026

PROVINCE OF ONTARIO

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EXPIRES: 01/31/2027

SAFETY CLASSIFICATION CODE

SC1

APPROVALS

DATE

DRAWN

10/10/2024

RESPONSIBLE ENGINEER

11/01/2024

SHEET TITLE

COVER SHEET - GENERAL NOTES & DETAILS

ISSUING AUTHORITY

CA-0050155

REVISION DATE

7/15/2026

REV

11

SHEET ID

G100

GE VERNOVA

HITACHI

ARRANGEMENT DRAWING

BWRX-300 REACTOR BUILDING

DP-SC DRAWING GENERAL NOTES

DRAWING NO

008N9536

REVISION DATE

7/15/2026

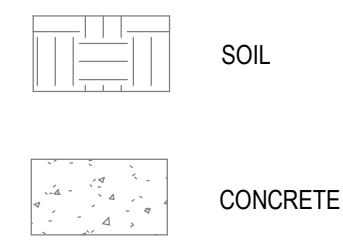
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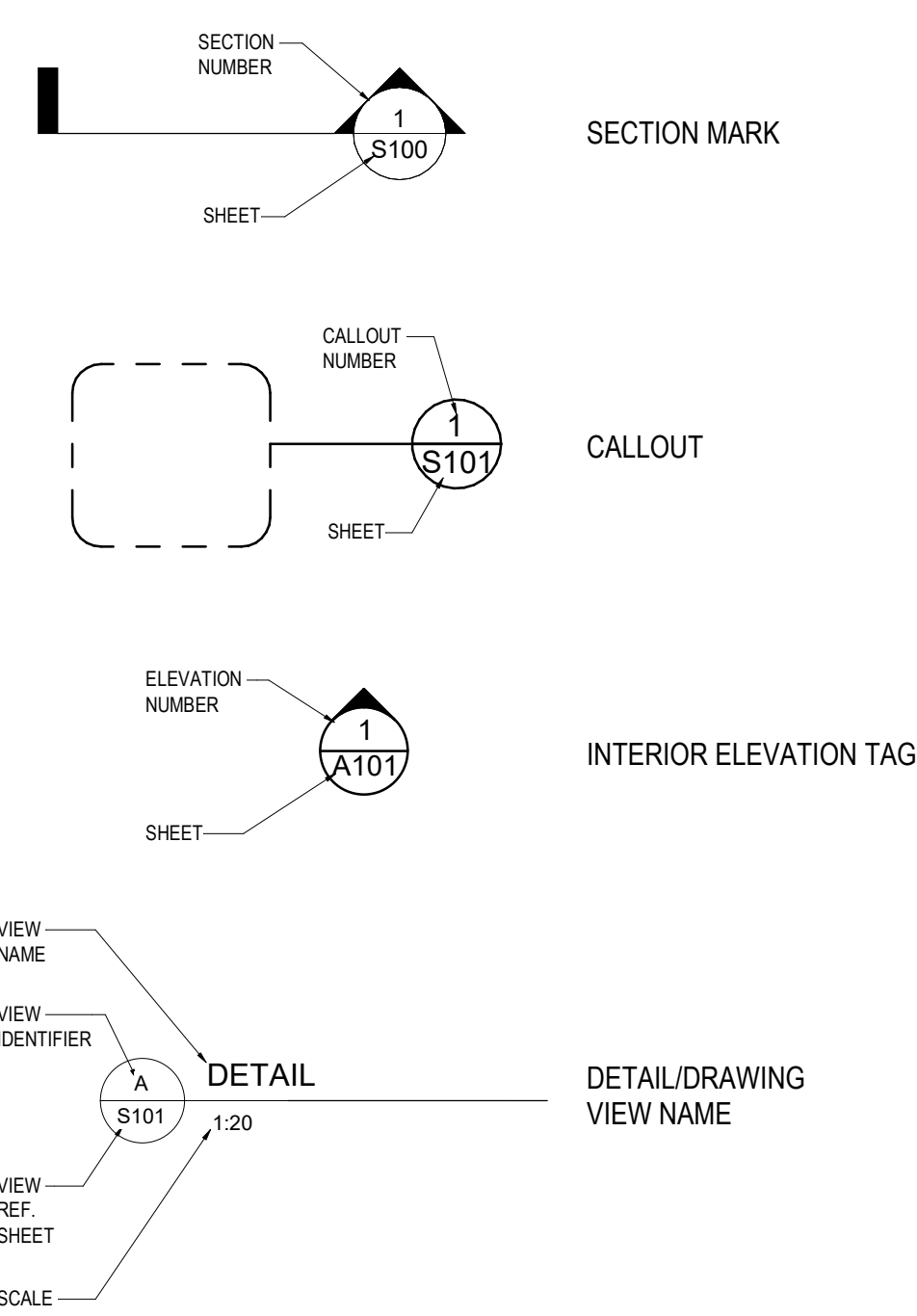
ABBREVIATIONS

& @ € Ø +/- SQ	AND AT CENTERLINE DIAMETER/ROUND PLUS OR MINUS SQUARE
BLDG BTM BTWN BRKT	BUILDING BOTTOM BETWEEN BRACKET
COL CONC CONN CONT CJP	COLUMNS CONCRETE CONNECTION CONTINUOUS COMPLETE JOINT PENETRATION
DIAPH DP-SC	DIAPHRAGM DIAPHRAGM PLATE STEEL-PLATE COMPOSITE
E EL ELEV EQ EXT	EAST ELEVATION ELEVATOR EQUAL EXTERIOR
FL FP	FLOOR FACEPLATE
HOR	HORIZONTAL
IN IS INTMD	INCH INSIDE INTERMEDIATE
LONG.	LONGITUDINAL
MAX MIN MISC MM	MAXIMUM MINIMUM MISCELLANEOUS MILLIMETER
N NTS NSSC	NORTH NOT TO SCALE NON-STRUCTURAL STEEL COMPOSITE
OPNG OPP OD OS OSW	OPENING OPPOSITE OUTSIDE DIAMETER OUTSIDE OUTER SHAFT WALL
PEN PERP PL PJP	PENETRATION PERPENDICULAR PLATE PARTIAL JOINT PENETRATION
RAD RB RC REF REINF REQ RQMT RPV	RADIUS REACTOR BUILDING REINFORCED CONCRETE REFERENCE/REFER REINFORCED/REINFORCING REQUIRED REQUIREMENT REACTOR PRESSURE VESSEL
S SCCV SF SIM SL SOG SPOG SS STD STL STIF STRUC STSC	SOUTH STEEL-PLATE COMPOSITE CONTAINMENT VESSEL SQUARE FOOT SIMILAR SLOPE SLAB ON GRADE SPACING STAINLESS STEEL STANDARD STEEL STIFFENER STRUCTURE/STRUCTURAL STRUCTURAL STEEL-PLATE COMPOSITE
T&B THK TOS TYP	TOP AND BOTTOM THICK/THICKNESS TOP OF STEEL TYPICAL
U.N.O	UNLESS NOTED OTHERWISE
VAR	VARIES
W W/ W/O WW	WEST WITH WITHOUT WING WALL

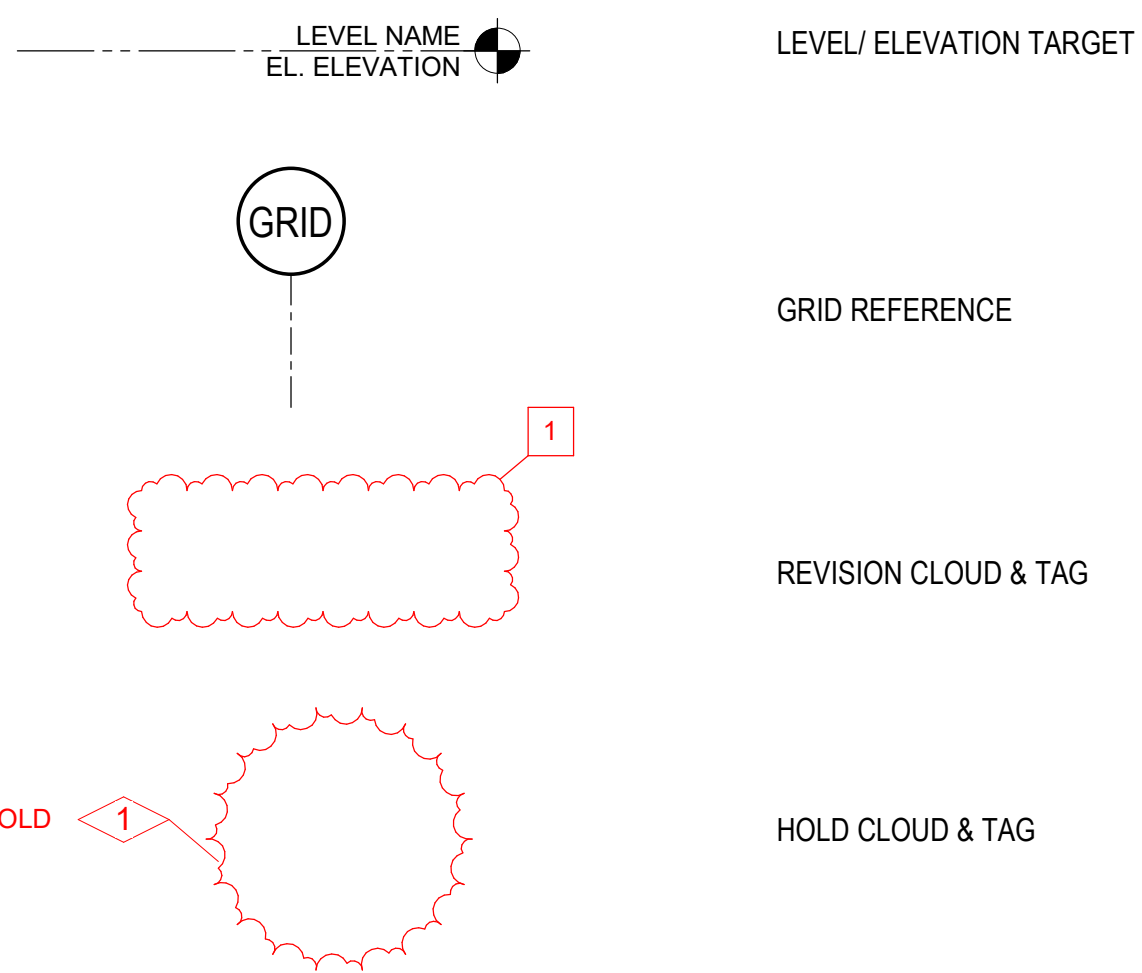
MATERIAL SYMBOLS



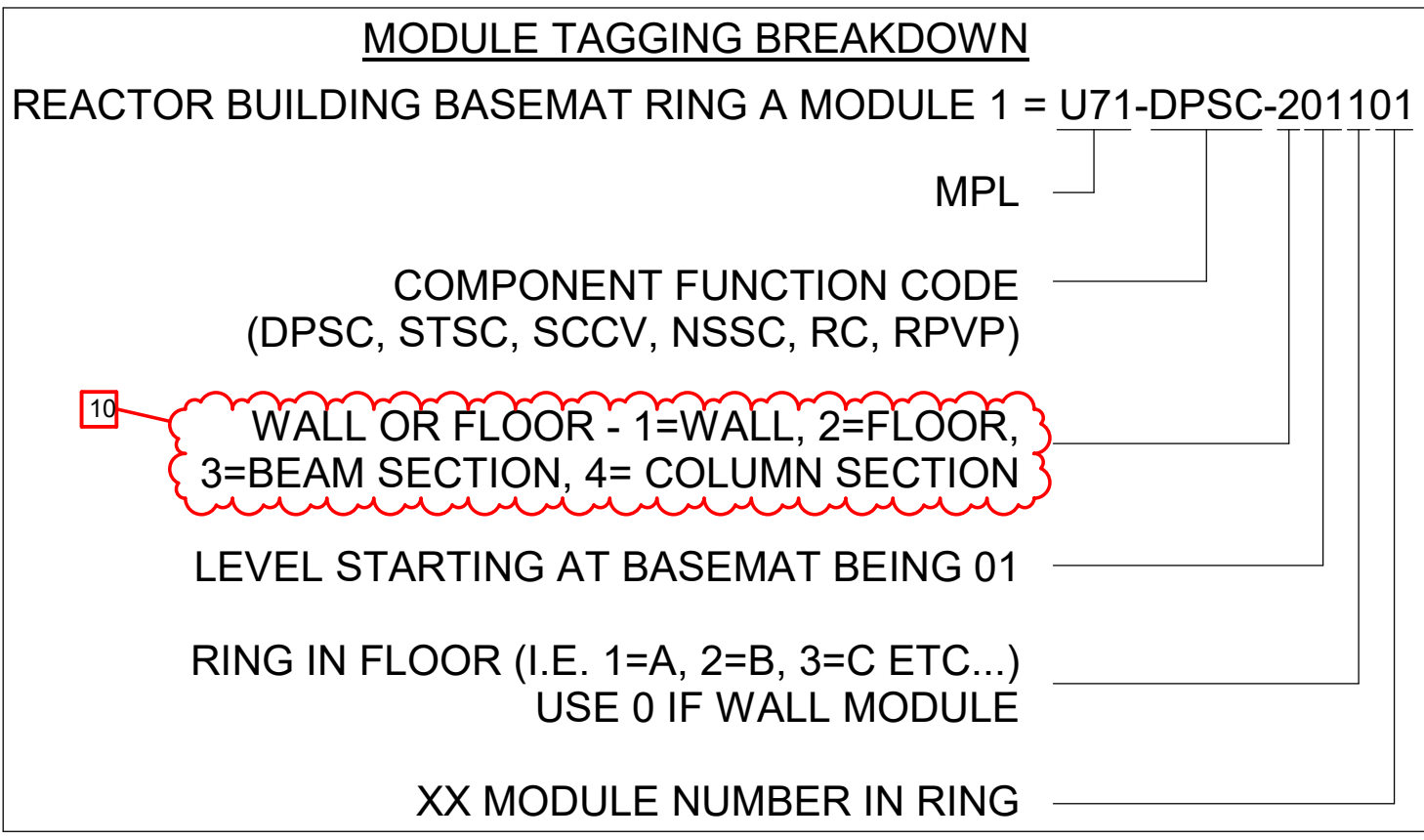
GRAPHIC SYMBOLS



ANNOTATION SYMBOLS



REVISIONS				
REV	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
7	09/02/2025	R LUECHT	CA-00068765 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI



- NOTES:
- ALL LINEAR UNITS ARE IN MILLIMETERS AND [FEET] UNLESS OTHERWISE NOTED. DUAL DIMENSIONS IN BRACKETS ARE FOR REFERENCE ONLY.
 - ANGULAR UNITS ARE IN DEGREES UNLESS OTHERWISE NOTED.
 - ELEVATION CALLOUTS ARE BASED ON TOP OF STEEL FACEPLATES AND REFERENCES TO LEVEL 0M. FOR FLOORS WHERE TOP OF STEEL FACEPLATE ELEVATION VARIES, ELEVATION REFERS TO TOP OF LOWER FACEPLATE.

ISSUED FOR CONSTRUCTION
WITH HOLDS
04/06/2026

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GENERAL NOTES

DRAWING NO.
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REV
10

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REVISIONS				
REV	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
9	11/21/2025	R LUECHT	CA-0061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-0063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI
11	7/15/2026	R.P. LUECHT	CA-00066870 SEE SHEET G100 FOR REVISION SUMMARY	P.K.OSTROWSKI

TABLE 1-5																		
Structural DP-SC Component	Location		Member Thickness t_m		Faceplate and Diaphragm Plate Thickness t_f		Steel Plate Grade	Filler Metal Classification Strength, X_m		Double Fillet Weld (Option 1)		PJP Weld + Fillet Reinforcement (Option 2)		PJP Weld + Fillet Reinforcement with single fillet weld on the plate backside (Option 3)				CJP Weld (Option 4.0)
										Fillet Weld Leg Size, t_{fil}		Required weld effective throat, (S)		Required weld effective throat (S)		Fillet Weld Leg Size t_{fil}		
	From Elevation	To Elevation	(in.)	(mm)	(in.)	(mm)		(ksi)	(Mpa)			(in.)	(mm)	(in.)	(mm)	(in.)	(mm)	
Inner Basemat	EI -35.2m [-115.6']	EI -34.0m [-114.6']	48	1220	0.5	12.7	ASME SA537 Class 2	80	551.6	0.375	9.5	0.625	15.9	0.5	12.7	0.375	9.5	Note (3, 4)
Outer Basemat	EI -35.2m [-115.6']	EI -34.0m [-114.6']	48	1220	0.5	12.7	ASTM A572 Grade 65	80	551.6	0.394	10	0.625	15.9	0.5	12.7	0.375	9.5	
SCCV Wall	EI -35.2m [-115.6']	EI -31.1m [-102.0']	48	1220	.75	19.1	ASME SA537 Class 2	80	551.6	0.625	15.9	0.875	22.2	0.75	19.1	0.625	15.9	
	EI -31.1m [-102.0']	EI -27.6m [-90.7']										1.250	31.8					
RB Outer Wall	EI -35.2m [-115.6']	EI -32.9m [-108.2']	36	915	.75	19.1	ASTM A572 Grade 65	80	551.6	0.625	15.9	0.875	22.2	0.75	19.1	0.625	15.9	
	EI -32.9m [-108.2']	EI -23.9m [-78.6']										1.125	28.5					
RPV Pedestal Wall	EI -34.0m [-114.6']	EI -25.5 [-83.5']	48	1220	.75 ⁽¹⁾	19.1 ⁽¹⁾	ASTM A572 Grade 50	70	482.6	0.625	15.9	1.125	28.5	0.75	19.1	0.625	15.9	
Interior Floor Slabs	EI -29.0m [-95.2']	EI -29.0m [-95.2']	24	609.6	0.5	12.7	ASTM A572 Grade 65	80	551.6	0.394	10.0	0.625	15.9	0.50	12.7	0.375	9.5	

Note:
(1) Faceplate thickness is 38.1mm [1 1/2"] for the pedestal first ring extending up to 610mm [2'-0"] minimum.
(2) Weld Option 3 can be used at the diaphragm plate ends in combination with weld option 2 in the middle zone to control diaphragm plate distortion. The length of Option 3.0 weld at the plate ends can be specified by the fabricator to control diaphragm plate distortion. However, plate preparation (groove angle and root face dimensions) should be the same for both options along the full plate length to avoid any stress concentrations.
(3) Fillet reinforcement to be used as required by NEDC-33926P Table 6-2 for Category G.
(4) CJP and PJP weld veld shape, angle, root opening, root face dimension, and reinforcing fillet weld dimensions shall be specified by the fabricator following welding code (listed in Table 1-8) requirements to be reviewed and accepted by Engineer as part of fabrication drawings review.

TABLE 1-7															
Structural Component	Member Thickness t_m		Faceplate and Diaphragm Plate Grade	MAXIMUM Diaphragm Plate Spacing W_m		Stud Anchor Material	Stud Anchor Length, l_a		Stud Anchor Diameter, ϕ_a		Transversal Studs/or Diaphragm Spacings s_t		Longitudinal Studs Spacings s_l		Number of Stud rows, n
	(in.)	(mm)		(in.)	(mm)		(in.)	(mm)	(in.)	(mm)	(in.)	(mm)			
Inner Basemat	48	1220	ASME SA537 Class 2	24	609.6	ASTM A108 Type B	6.188	157.2	0.750	19.1	Wsc3	203.2	10.0	254.0	2.00
				20	508.0						Wsc2	254.0	6.0	152.4	1.00
				<10	<254.0						No Studs				
Outer Basemat	48	1220	ASTM A572 Grade 65	24	609.6	ASTM A108 Type B	6.188	157.2	0.750	19.1	Wsc3	203.2	10.0	254.0	2.00
				20	508.0						Wsc2	254.0	6.0	152.4	1.00
				<10	<254.0						No Studs				
				24	609.6						Wsc3	203.2	3.0	76.2	2.00
				16	406.4						Wsc2	203.2	3.0	76.2	1.00
Interior Floor Slabs from -29.0m to -8.5m	24	609.6		6	152.4										
			No Studs												

NOTES:

(1) These diaphragm plates spacings are presented for DP-SC basemat representing the stud requirements at the maximum allowed diaphragm's spacing and the reduced studs as the diaphragm's spacing decrease

(2) Minimum stud spacing is 76.2 mm [3"] for 19.05 mm [3/4"] stud diameter per AISC 360 Sect. 18.3c. Maximum spacing between studs or stud to diaphragm is 254mm [10"] to prevent buckling.

(3) For radially arranged diaphragms where the diaphragm spacings vary along the middle line, the Transversal Studs Spacings can be reduced between the middle rows to provide 152.4mm [6"] clearance between the diaphragms and first stud rows

(4) Shifting of studs is permitted where necessary to avoid weld lines or interference, as long as the maximum/minimum spacing requirements are met between stud center to other structure.

(5) Minimum spacing from the edge of a stud base to the free edge of a plate or concrete floor hole shall not be less than 38.1mm [1.5"]

(6) In the case of double stud rows within a diaphragm cavity floor module, longitudinal spacing for studs in interior floor slabs can be extended to 88.9 mm [3.5"] center to center where needed to straddle thickened plate welded steel seams.

(7) In the case of double stud rows within a diaphragm cavity floor module with $W_m = 610mm$ [24"] where studs cannot be located in the stud array at the prescribed longitudinal spacing, studs shall be added within the affected region, to restore the stud density for that area to a minimum density of 8 studs within a 305mm [12"] longitudinal length of diaphragm channel in the area of the skipped studs.

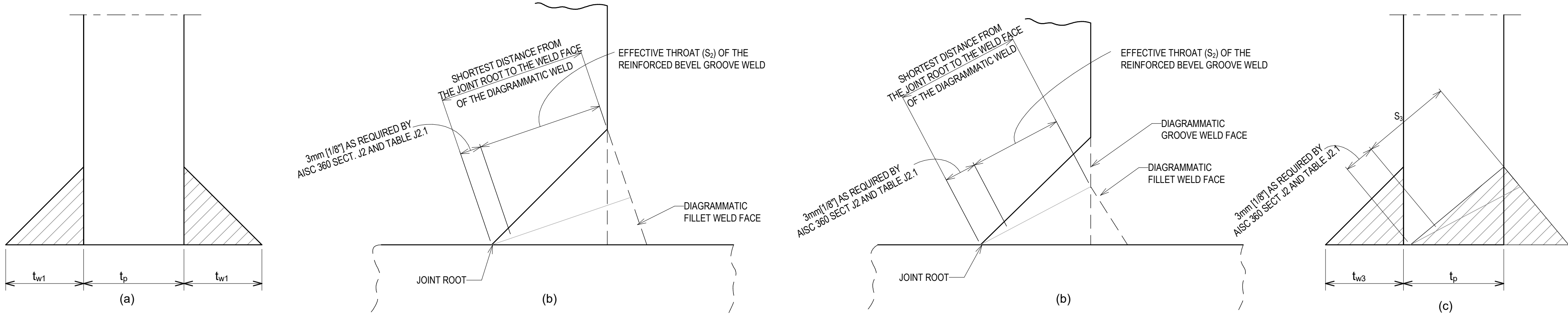
(8) In the case of single stud rows within a diaphragm cavity floor module with $W_m = 406mm$ [16"] where studs cannot be located in the stud array at the prescribed longitudinal spacing, studs shall be added within the affected region, to restore the stud density for that area to a minimum density of 4 studs within a 305mm [12"] longitudinal length of diaphragm channel in the area of the skipped studs.

(9) Stud-to-sleeve spacing can be up to 5". Stud density requirements specified in Table 1-7 shall apply to all areas outside the sleeve region.

(10) The minimum distance from edge of stud base to the cover plate or sleeve is 38.1mm [1.5"]

(11) Longitudinal stud spacing (S) of 122 mm [4.8"] between floor rings C and B at the interior floor slab ring splice, Level -29.0 m, is acceptable.

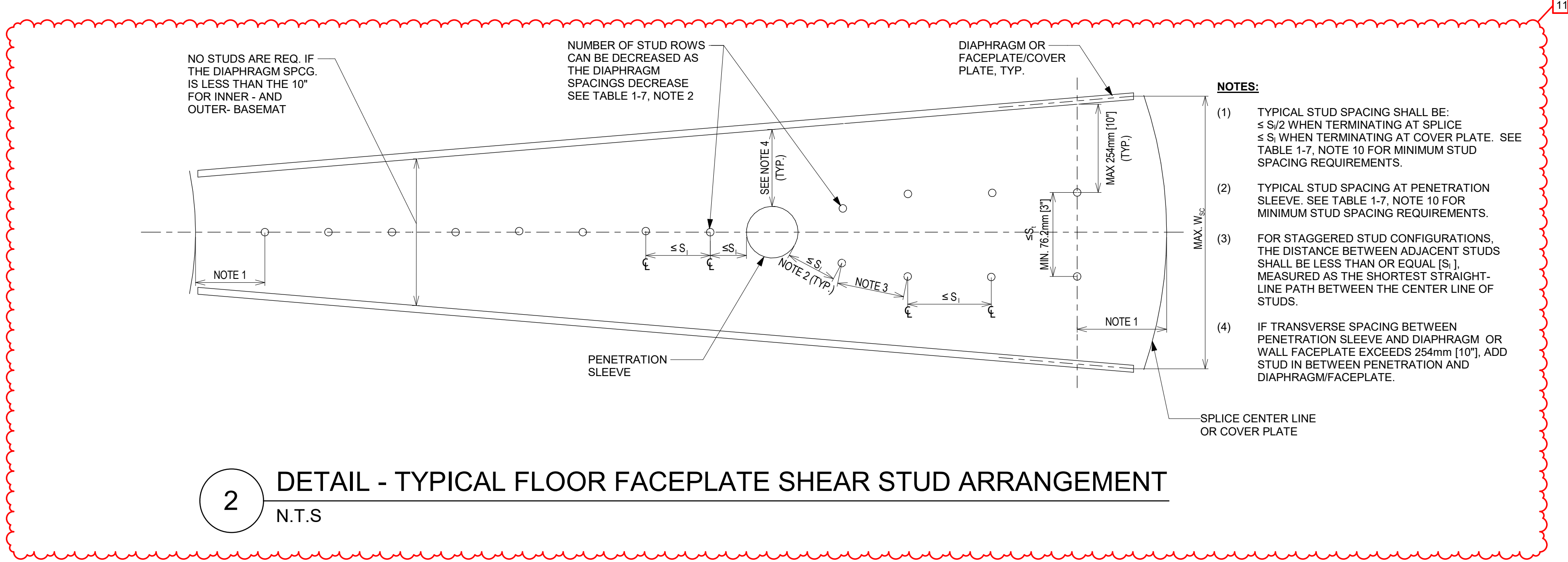
NOTES:
(1) Three diaphragm plates spacings are presented for DP-SC basemat representing the stud requirements at the maximum allowed diaphragm's spacing and the reduced studs as the diaphragm's spacing decrease.
(2) Minimum stud spacing is 76.2 mm [3"] for 19.05 mm [3/4"] stud diameter per AISC 360 Sect. 18.3e. Maximum spacing between studs or stud to diaphragm is 254mm [10"] to prevent faceplate buckling.
(3) For radially arranged diaphragms where the diaphragm spacings vary along the module length, the transversal Studs Spacings can be reduced between the middle rows to provide 152.4mm [6"] clearance between the diaphragm plates and first stud rows.
(4) Shifting of studs is permitted where necessary to avoid weld lines or interference, as long as the maximum/minimum spacing requirements are met unless noted otherwise.
(5) Minimum spacing from the edge of a stud base to the free edge of a plate or concrete flow hole shall not be less than 38.1mm [1 1/2"].
(6) In the case of double stud rows within a diaphragm cavity floor module, longitudinal spacing for studs in interior floor slabs can be extended to 88.9 mm [3.5"] center to center where needed to straddle thick/thin plate transition weld seams.
(7) In the case of double stud rows within a diaphragm cavity floor module with $W_{sc} = 610mm$ [24"] where stud(s) cannot be located in the stud array at the prescribed longitudinal spacing, studs shall be added within the affected region, to restore the stud density for that area to a minimum density of 8 studs within a 305mm [12"] longitudinal length of diaphragm channel in the area of the skipped studs.
(8) In the case of single stud rows within a diaphragm cavity floor module with $W_{sc} = 406mm$ [16"] where stud(s) cannot be located in the stud array at the prescribed longitudinal spacing, studs shall be added within the affected region, to restore the stud density for that area to a minimum density of 4 studs within a 305mm [12"] longitudinal length of diaphragm channel in the area of the skipped studs.
(9) Stud-to-sleeve spacing can be up to S_t . Stud density requirements specified in table 1-7 shall apply to all areas outside the sleeve region.
(10) The minimum distance from edge of stud base to the cover plate or sleeve is 38.1mm [1 1/2"].
(11) Longitudinal stud spacing (S_l) of 122 mm [4.8"] between floor rings C and B at the interior floor slab ring splice, Level -29.0 m, is acceptable.



DP-SC DIAPHRAGM PLATE TO FACEPLATE WELD:
(a) OPTION 1 DOUBLE FILLET WELD, (b) OPTION 2 PJP PLUS FILLET REINFORCEMENT, AND (c) OPTION 3 PJP WELD + FILLET REINFORCEMENT WITH A SINGLE FILLET WELD ON THE PLATE BACK SIDE.

1 DETAIL - TYPICAL WELD JOINT CONFIGURATION

N.T.S

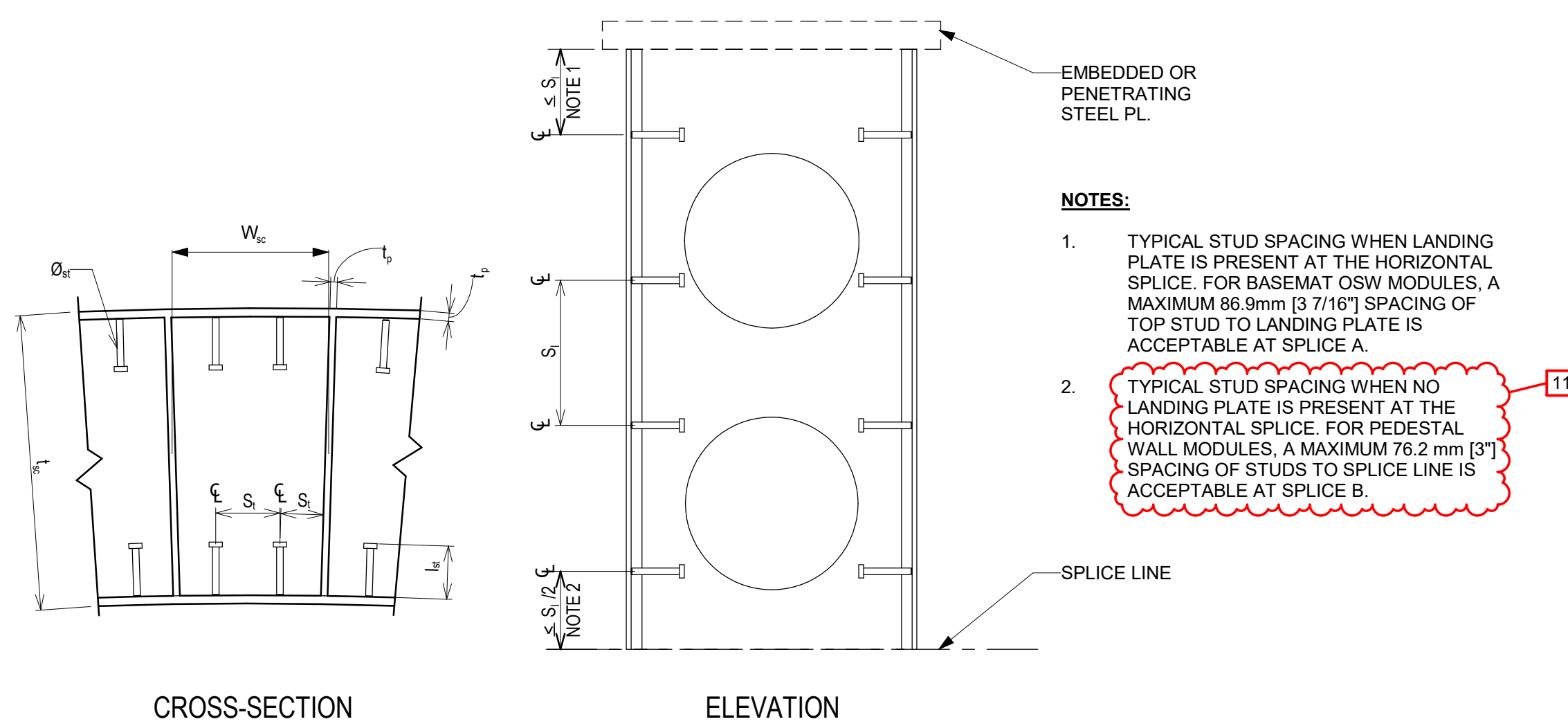


2 DETAIL - TYPICAL FLOOR FACEPLATE SHEAR STUD ARRANGEMENT

N.T.S

TABLE 1-6																	
Structural Component	Location	Member Thickness t_m		Faceplate and Diaphragm Plate Grade	Maximum Diaphragm Plate Spacing @ the Member Centerline W_m		Stud Anchor Material	Stud Anchor Length, l_a		Stud Anchor Diameter, ϕ_a		Number of Stud Columns	Maximum Longitudinal Studs Spacings (or stud to plate)		Transversal Studs Spacings s_t	Maximum transversal studs spacing (or stud to diaphragm)	
		(in.)	(mm)		(in.)	(mm)		(in.)	(mm)	(in.)	(mm)		(in.)	(mm)		(in.)	(mm)
SCCV Wall	From EL. -35.2m [-115.6'] to EL. -25.5m [-83.5']	48	1220	ASME SA537 Class 2	24	609.6	ASTM A108 Type B	6.316"	157.2	3/4"	19.1	2	4.0	101.6	Wsc3	16	406.4
RB Outer Wall	From EL. -35.2m [-115.6'] to EL. -23.9m [-78.6']	36	915	ASTM A572 Grade 65	18.0	457.2	ASTM A108 Type B	6.316"	157.2	3/4"	19.1	1	3.0	76.2	Wsc2	15	381.0
RPV Pedestal Wall	From EL. -34m [-114.6'] to EL. -31.1m [-102']	48	1220	ASTM A572 Grade 50	24	609.6	ASTM A108 Type B	6.316"	157.2	3/4"	19.1	2	4.0	101.6	Wsc3	17	431.8
	From EL. -31.1m [-114.6'] to EL. -8.0m [-26.2']												8	203.2			

NOTES:
(1) Minimum studs spacing is 3 in (76.2 mm) per AISC 360 Sect. 18.3e.
(2) Shifting of studs is permitted where necessary to avoid weld lines or interference, as long as the maximum/minimum spacing requirements are met unless noted otherwise.
(3) Minimum spacing from the edge of a stud base to the free edge of a plate or concrete flow hole or penetration sleeve edge shall not be less than 38.1mm [1 1/2"].
(4) Where penetration sleeve is located within shaft walls, maintain the required stud density outside the sleeve region. Add additional studs as required to ensure Transversal Stud Spacing between sleeve and adjacent diaphragm does not exceed the specified Maximum Transversal Stud Spacing.



3 DETAIL - TYPICAL WALL FACEPLATE SHEAR STUD ARRANGEMENT

N.T.S

TABLE 1-8						
Component ⁽¹⁾	Location	Material Type	Lowest Anticipated Service Temperature ⁽²⁾	Welding Code/Standard ^(2*)	Welding Inspection Requirements	Static/Cyclic Loading
Inner Basemat (Containment) ⁽³⁾	From EL. -35.2m [-115.6'] to EL. -34m [-114.6']	ASME SA 537 CLASS 2	0°C [32°F]	ASME Section IX	NEDC-33926P Table 6-2	Statically Loaded
Outer Basemat (non-Containment)	From EL. -35.2m [-115.6'] to EL. -34m [-114.6']	ASTM A572 GRADE 65	0°C [32°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 10% inspected by ultrasonic examination or radiographic examination	Statically Loaded
SCCV Wall ⁽³⁾	From EL. -35.2m [-115.6'] to EL. 0.0m [0.0']	ASME SA 537 CLASS 2	10°C [50°F]	ASME Section IX	NEDC-33926P Table 6-2	Statically Loaded
RB Outer Wall	From EL. -35.2m [-115.6'] to EL. 0.0m [0.0']	ASTM A572 GRADE 65	0°C [32°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 10% inspected by ultrasonic examination or radiographic examination. UNO on the detail drawings	Statically Loaded
RPV Pedestal Wall	From EL. -34m [-114.6'] to EL. 0.0m [0.0']	ASTM A572 GRADE 50	10°C [50°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 100% inspected by ultrasonic examination or radiographic examination. UNO on the detail drawings	Statically Loaded
RPV Pedestal Attachments	From EL. -34m [-114.6'] to EL. 0.0m [0.0']	ASTM A572 GRADE 50	10°C [50°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 100% inspected by ultrasonic examination or radiographic examination. UNO on the detail drawings	Statically Loaded
Interior Floor Slabs	EL. 0.0m [0.0']	ASTM A572 GRADE 65	10°C [50°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 10% inspected by ultrasonic examination or radiographic examination. UNO on the detail drawings	Statically Loaded
Wing Walls, Stair A & B Walls, and Elevator Walls	From EL. -34m [-114.6'] to EL. 0.0m [0.0']	ASTM A572 GRADE 50	10°C [50°F]	AWS D1.1 / D1.1 M or ASME Section IX	ANSIAISC N690 Sect. NN5. CJP welded joints shall be 10% inspected by ultrasonic examination or radiographic examination. UNO on the detail drawings	Statically Loaded

NOTES:
(1) ANSIAISC N690 Sect. NA3, additional requirements for components subjected to Impactive and/or Impulsive are not applicable to DP-SC components below elevation -29.0m. For DPSC components at EL -29.0m or above it can be conservatively assumed that impactive and/or impulsive loads exist or to be confirmed by engineering on a case-by-case basis.
(2) Minimum temperature of outer and inner faceplates under winter operating conditions is considered. For ground temperature during winter 0°C [32°F] is conservatively assumed.
(3) All ferritic materials including base metal, welding filler materials and base metal heat affected zone, shall be impact tested in accordance with the requirements of the ASME Code, Section III, Subarticles CC-2500, NE-2300, and NF-2300 as applicable. Steel plates having thickness that exceeds 6 mm [1/4"] shall be normalized or quenched and tempered, fully killed, and melted to a fine grain melting practice, whenever the material is not impact tested. See Note 6 for filler material...
(4) As per ANSIAISC 360 Sect. J2.7 when Charpy V-notch (CVN) toughness is specified, the process consumables for all weld metal, tack welds, root pass, and subsequent passes deposited in a joint shall be compatible to ensure notch-tough composite weld metal.
(5) Filler metal with a specified minimum Charpy V-notch (CVN) toughness of 20 ft-lb (27 J) at 40°F (4°C) or lower shall be used for CJP welded joints per ANSIAISC NJ 6 and ASME 360-22 J2.6.
(6) For the purpose of Welding Procedure Specification (WPS) and Performance Qualification of Welding Personnel, ASME Section IX can be used in lieu of AWS D1.1 / D1.1 M as permitted by AWS D1.1 / D1.1 M Clauses 6.2.1.2 and 6.2.2.1.
(7) Welds to SCCV shall follow NEDC-33926P Section 6.0 requirements.
(8) Material for floor bracket is ASTM A572 GRADE 65 from interior floor slabs to SCCV shaft wall connection components.



APPROVALS		DATE
DRAWN	R. P. LUECHT	10/10/2024
REVIEWED	OSAMA BEDAIR	11/01/2024
SHEET TITLE		
GENERAL WELDING AND STUD TABLES		

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008N9536	7/15/2026	11
ISSUING AUTHORITY	SHEET ID	
CA-0050155	G103	

REVISIONS				
REV	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI
11	7/15/2026	R.P. LUECHT	CA-00066870 SEE SHEET G100 FOR REVISION SUMMARY	P.K.OSTROWSKI

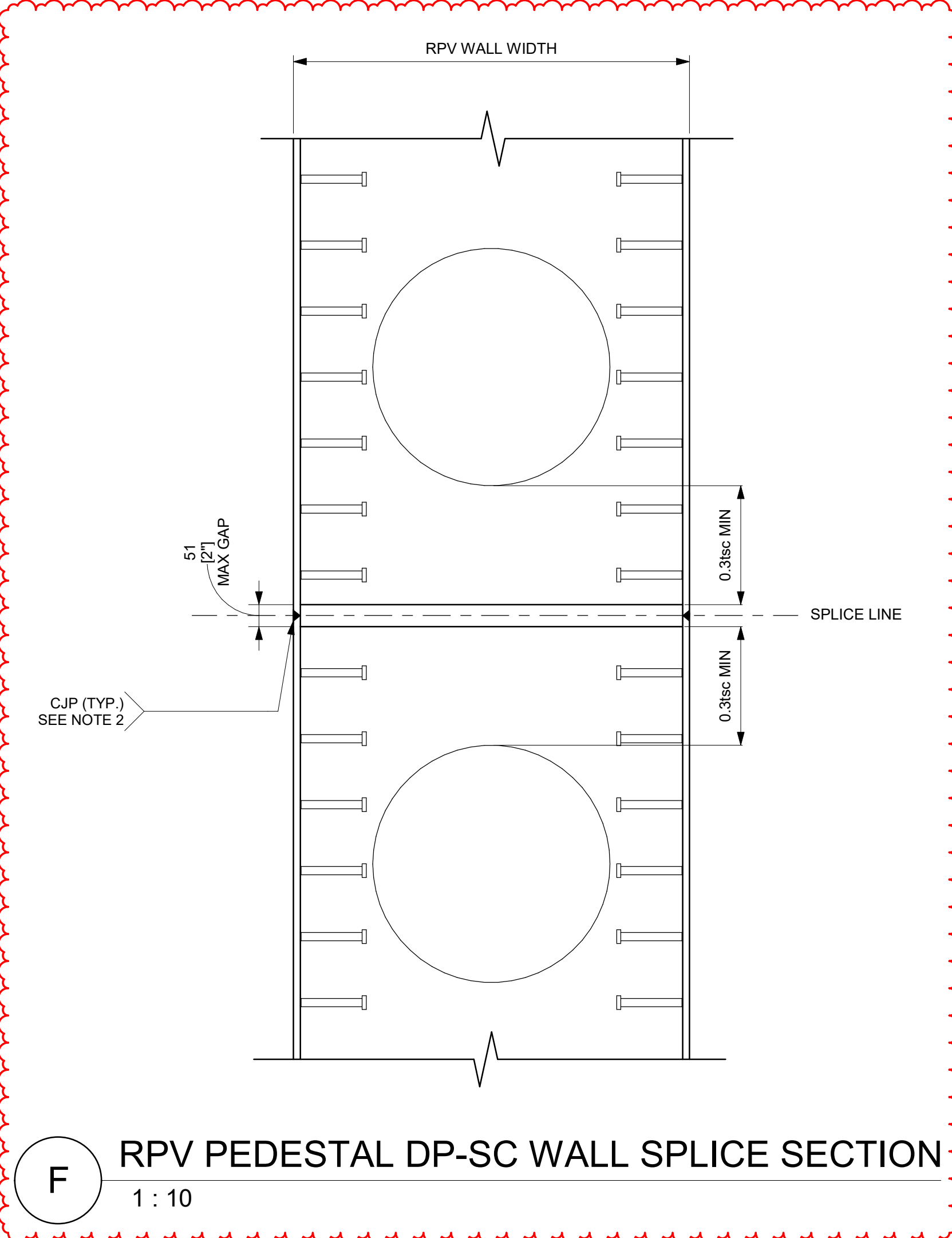
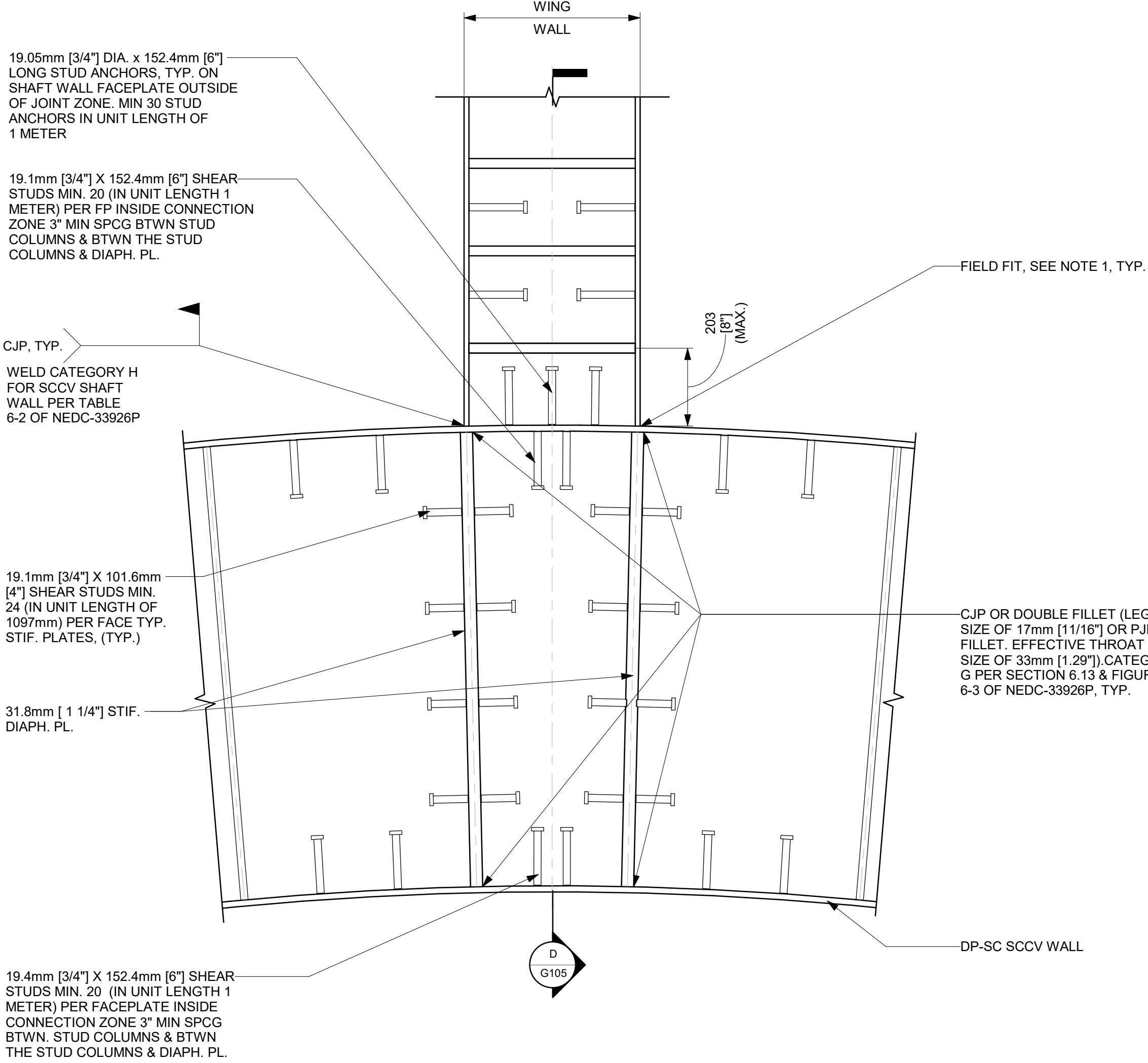
- NOTES:**
- THE ENDS OF THE SC WALL MODULE FACEPLATES (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) SHALL BE TRIMMED AND DIRECTLY ATTACHED TO THE FACEPLATE SURFACE OF THE SHAFT WALL MODULE. AT THE CONSTRUCTOR'S DISCRETION, A PATCH PLATE CAN BE USED TO EXTEND THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) WHEN A GAP IS PRESENT ON SITE. THE PATCH PLATE'S THICKNESS AND MATERIAL SHALL BE IDENTICAL TO THE SC WALL'S FACEPLATE AND SHALL BE CONNECTED TO THE FACEPLATE USING A CJP WELD.
 - CJP CRITICAL WELDED JOINT TO BE MINIMUM 50% INSPECTED BY ULTRASONIC OR RADIOGRAPHIC EXAMINATION. FOR RPV PEDESTAL SPLICE BETWEEN AND INCLUDING EL -34.0M AND EL -28.3M.

TABLE 1-9													
STRUCTURAL COMPONENT	MEMBER DEPTH (t.)		FACEPLATE & DIAPHRAGM PLATE THICKNESS (t.)		STEEL PLATE GRADE	F _t		F _y		THICKENED FACEPLATE THICKNESS "T"		LENGTH OF THICKENED PLATE "X"	
	(in.)	(mm)	(in.)	(mm)		(ksi)	(MPa)	(ksi)	(MPa)	(in.)	(mm)	(in.)	(mm)
INNER BASEMAT (SCCV)	48	1220	0.5	12.7	ASME SA537 CLASS 2	60	413.7	80	551.6	1.25	31.8	24	610
OUTER BASEMAT	48	1220	0.5	12.7	ASTM A572 GRADE 65	65	448.2	80	551.6	1.25	31.8	24	610

B SPLICE DETAIL - FLOOR MODULES
1 : 20

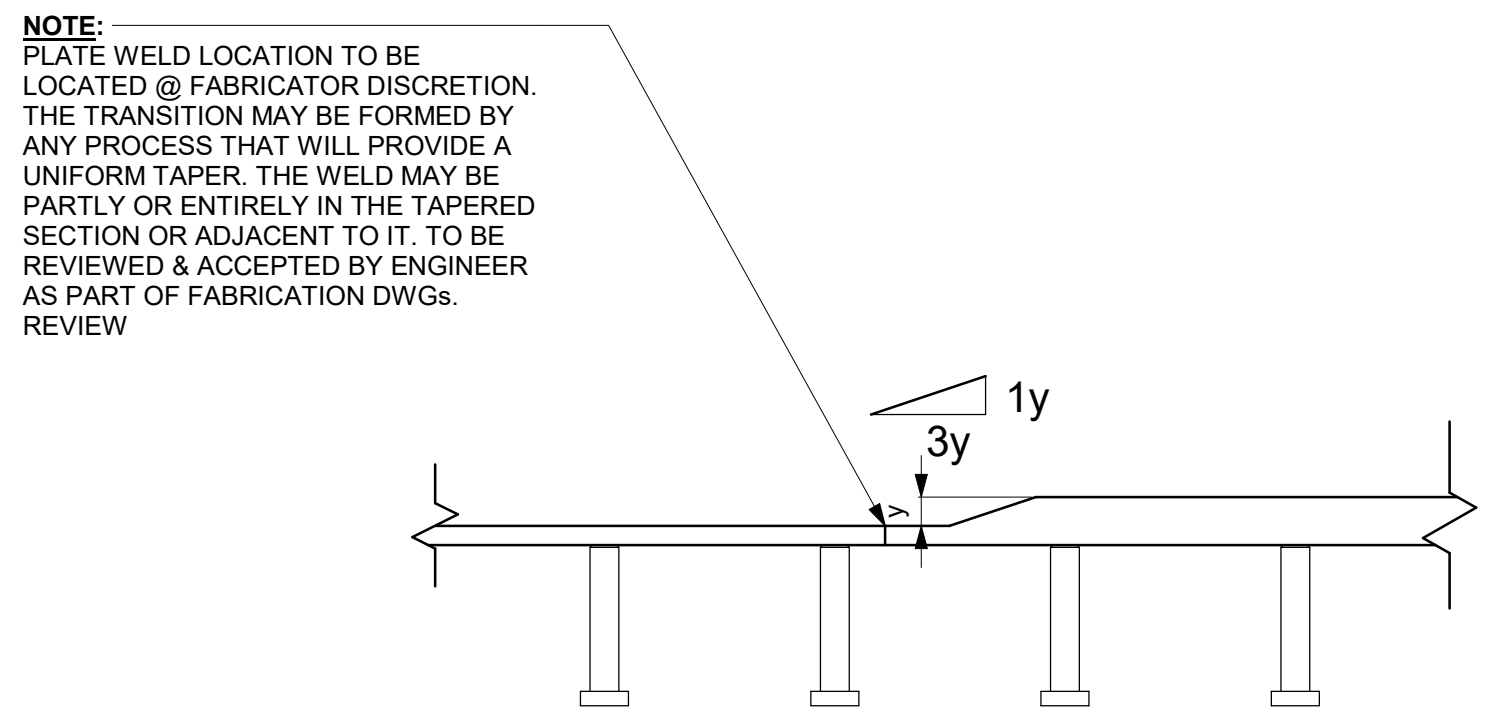
C SPLICE DETAIL - WALL MODULES
1 : 20

A SPLICE DETAIL - BASEMAT TO BASEMAT
1 : 20



D DETAIL - WING AND STAIR WALL CONNECTION @ RB WALL
1 : 10

E DETAIL - WING WALL CONNECTION @ SCCV WALL
1 : 10



H SPLICE DETAIL - FLOOR TO FLOOR
1 : 20

TABLE 1-10													
STRUCTURAL COMPONENT	MEMBER DEPTH (t.)		FACEPLATE & DIAPHRAGM PLATE THICKNESS (t.)		STEEL PLATE GRADE	F _t		F _y		THICKENED FACEPLATE THICKNESS "T"		LENGTH OF THICKENED PLATE "X"	
	(in.)	(mm)	(in.)	(mm)		(ksi)	(MPa)	(ksi)	(MPa)	(in.)	(mm)	(in.)	(mm)
INTERIOR FLOOR SLABS	24	610	0.5	12.7	ASTM A572 GRADE 65	65	448.2	80	551.6	1	31.8	12	305

ENGR: PETER OSTROWSKI
LICENSE: 90284142
EXPIRES: 01/31/2027

APPROVALS

R. P. LUECHT
RESPONSIBLE ENGINEER
OSAMA BEDAIR

DATE

10/10/2025
10/28/2025

STANDARD DETAILS

DRAWING NO.
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REVISION DATE
7/15/2026

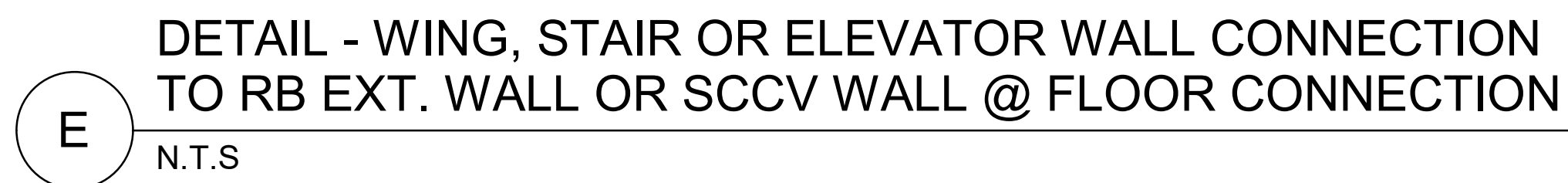
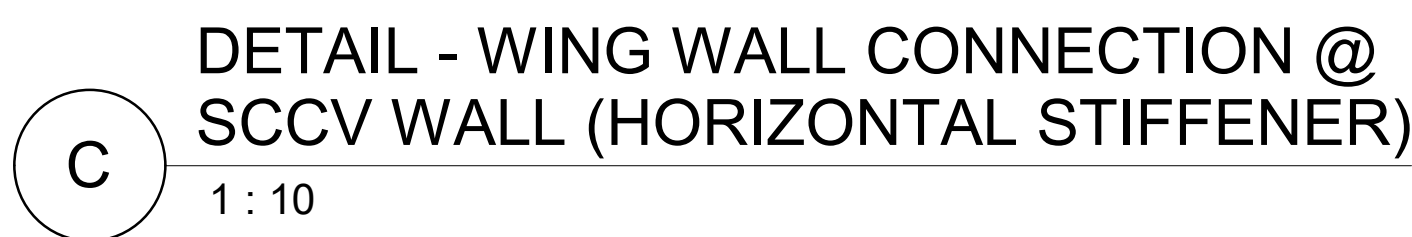
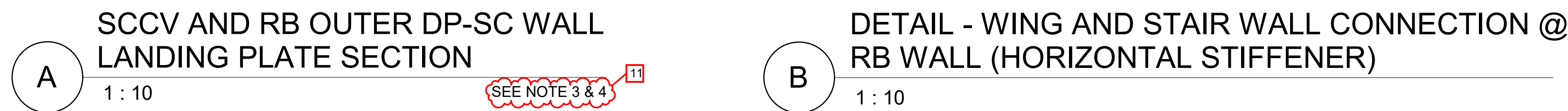
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1. THE ENDS OF THE SC WALL MODULE FACEPLATES (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) SHALL BE TRIMMED AND DIRECTLY ATTACHED TO THE FACEPLATE SURFACE OF THE SHAFT WALL MODULE. AT THE CONSTRUCTOR'S DISCRETION, A PATCH PLATE CAN BE USED TO EXTEND THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) WHEN A GAP IS PRESENT ON SITE. THE PATCH PLATE'S THICKNESS AND MATERIAL SHALL BE IDENTICAL TO THE SC WALL'S FACEPLATE AND SHALL BE CONNECTED TO THE FACEPLATE USING A CJP WELD.
2. FIELD FIT EACH DOUBLER PLATE SEGMENT SO THAT EDGE OF THE DOUBLER PLATE ALIGNS WITH CENTER OF RB WALL THICKENED DIAPHRAGM PLATE. IF THE DOUBLER PLATE CANNOT FULLY COVER THE ELEVATOR WALL'S FACEPLATE, THE EDGE OF THE DOUBLER PLATE CAN BE EXTENDED UP TO THE OUTER SURFACE OF THE THICKENED DIAPHRAGM PLATE. DOUBLER PLATE SEGMENTS NEED NOT TO BE ALIGNED FROM ONE WALL MODULE TO THE NEXT MODULE.
3. IF MINIMUM GAP SIZE OF 25.4mm [1"] CANNOT BE ATTAINED, 19mm [11/16"] MINIMUM CAN BE ACCEPTED WITH CONCURRENCE FROM ENGINEER.
4. FOR LANDING PLATE DETAILS, REFER TO DRAWING 011N0242 (SCCV) OR 011N0243 (EXTERIOR WALL).



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9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI
11	7/15/2026	R.P. LUECHT	CA-00066870 SEE SHEET G100 FOR REVISION SUMMARY	P.K.OSTROWSKI

CONNECTION NOTES:

- VERIFY ALL DIMENSIONS BEFORE STARTING WORK. THE A/E AND CONTRACTING OFFICER OF DESIGNATED REPRESENTATIVE SHALL BE NOTIFIED IMMEDIATELY OF ANY DISCREPANCIES FOUND.
- STIFFENER PLATE SHALL BE ASTM A572 GR 65.
- THE STUD ANCHOR WELDED ON THE STIFFENER CAN BE OMITTED IF IT CONFLICTS WITH ARCHITECTURAL OPENING SUCH AS DOOR OPENING.
- A GAP IS PROVIDED AT THE END OF FLOOR PANEL TO PREVENT THE HEADED STUDS WELDED ON THE SC WALL FACEPLATE FROM INTERFERING WITH FLOOR PANEL DURING FIELD INSTALLATION. THIS GAP WILL BE CLOSED WITH A SHORT SEGMENT OF FACEPLATE (IN-FILL FACEPLATE) TO FORM CONTINUITY OF THE FACEPLATE TO THE WALL, AFTER THE FLOOR PANEL IS IN PLACE.
- THE MAXIMUM RECOMMENDED SPACING BETWEEN THE WALL FACEPLATE AND THE FIRST DIAPHRAGM PLATE IS 305mm [12"].
- STUDS MINIMUM SPACING IS 76mm [3"] AND MAXIMUM SPACING IS 610mm [24"].
- IF A WALL VERTICAL SPLICE EXISTS BETWEEN TWO CONCRETE FLOW HOLES AND THE SPACING BETWEEN THE HOLES DEVIATES FROM THE SPECIFIED UNIT LENGTH (1 METER), ADJUST THE NUMBER OF SHEAR STUDS BETWEEN THE CONCRETE FLOW HOLES PROPORTIONALLY, BASED ON THE RATIO OF THE ACTUAL SPACING TO THE SPECIFIED UNIT LENGTH.
- FOR STIFFENER PLATES WITH RECTANGULAR FLOW HOLES, IT IS RECOMMENDED TO PROVIDE A CORNER RADIUS OF 2 TIMES THE PLATE THICKNESS, OR 16mm [5/8"], WHICHEVER IS GREATER, AT THE REINTRANT CORNERS. AT A MINIMUM, THE CORNER RADII OF AT LEAST 13mm [1/2"] SHALL BE PROVIDED.
- WELD BEFORE POURING CONCRETE INTO FLOOR DP-SC PANEL MODULE.
- IF THE CONCRETE FLOW IS LOCATED UNDER DOOR OPENING, THE CONCRETE FLOW HOLE MAY BE OMITTED AND TWO 19mm [3/4"] x 102mm [4"] SHEAR STUDS SHALL BE ADDED TO THE CENTER LINE OF THE OMITTED CONCRETE FLOW HOLE, UNDER THE STIFFENER PLATE.
- IF THE CONCRETE FLOW HOLE LOCATION CONFLICTS WITH TIE ROD POSITION- EITHER ABOVE OR BELOW- THE FLOW HOLE MAY BE RELOCATED WITHIN A TOLERANCE OF 6" TO MAINTAIN AN UNOBSTRUCTED CONCRETE FLOW PATH. THE NUMBER OF SHEAR STUDS BETWEEN ADJACENT CONCRETE FLOW HOLES SHALL BE ADJUSTED PROPORTIONALLY, BASED ON THE RATIO OF THE ACTUAL SPACING TO THE SPECIFIED UNIT LENGTH.
- THE ENDS OF THE SC WALL-TO-WALL CONNECTING FACEPLATES (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) SHALL BE TRIMMED AND DIRECTLY ATTACHED TO THE NEXT ADJACENT WALL MODULE AT THE CONSTRUCTOR'S DISCRETION. A PATCH PLATE CAN BE USED TO EXTEND THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) WHEN A GAP IS PRESENT ON SITE. THE PATCH PLATE'S THICKNESS AND MATERIAL SHALL BE IDENTICAL TO THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) AND SHALL BE CONNECTED TO THE FACEPLATE USING A CJP WELD.

DETAIL - FLOOR TO WING WALL CONNECTION (TYPE-1)

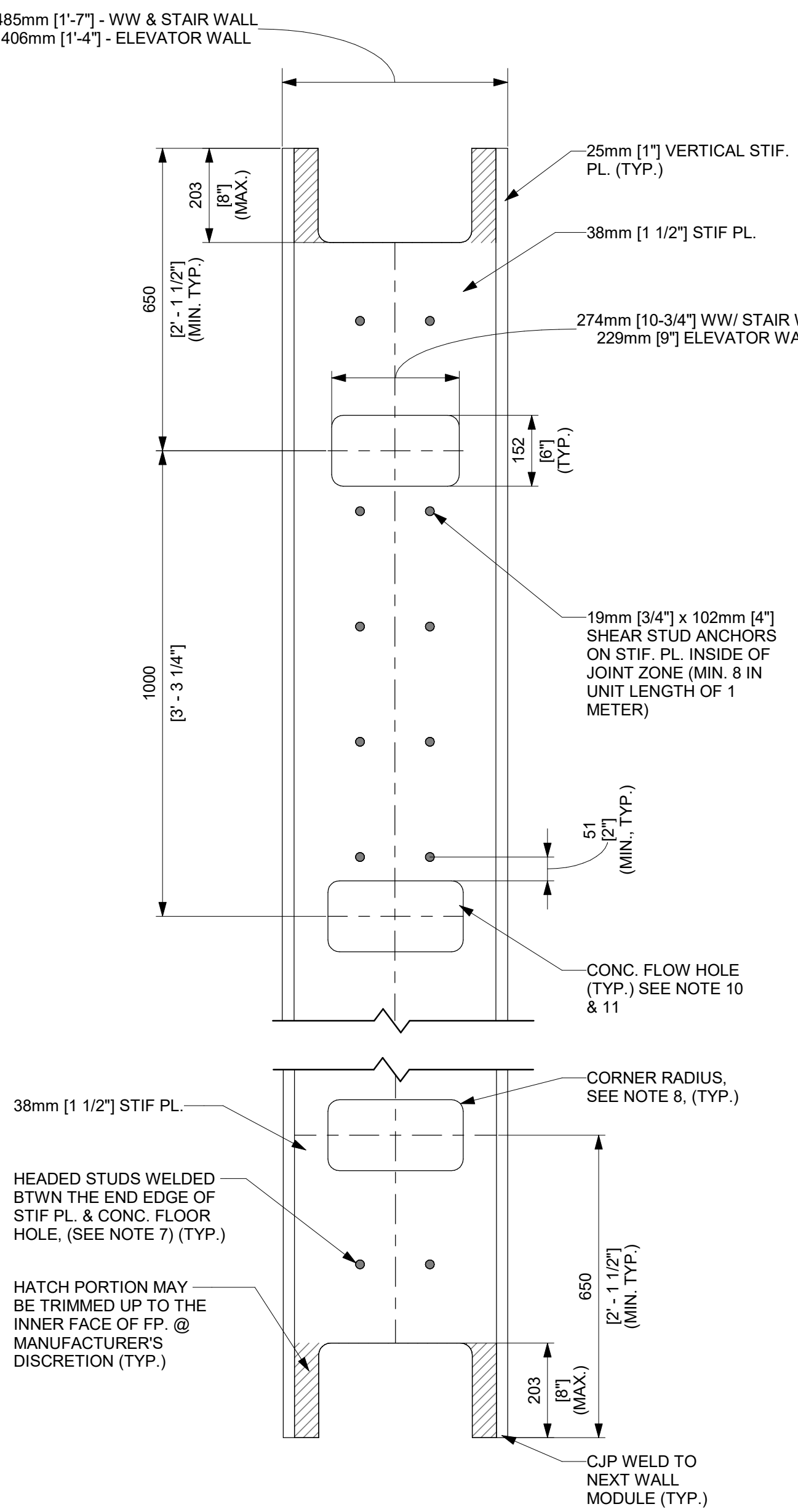
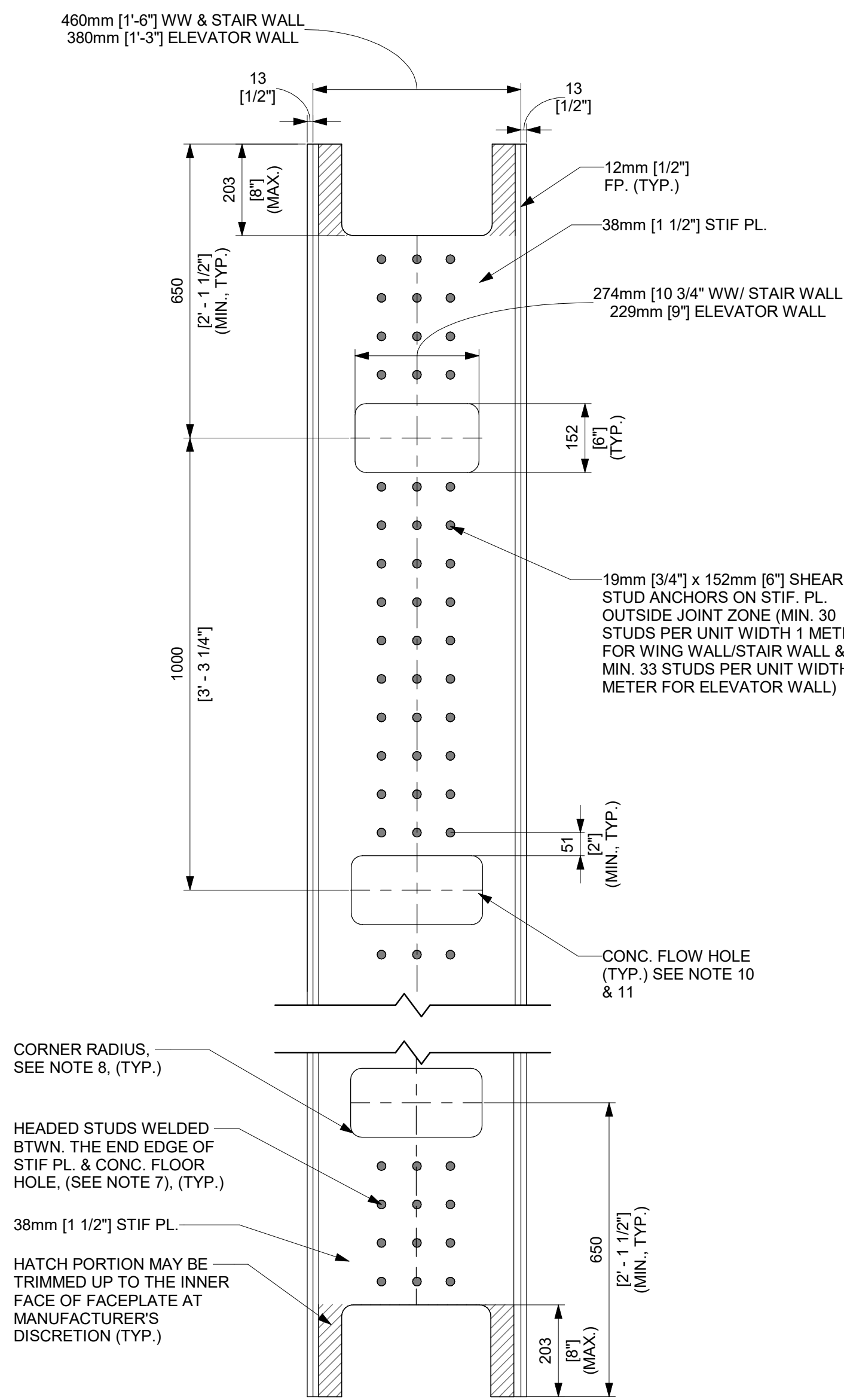
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DETAIL - FLOOR TO STAIR & ELEVATOR WALL CONNECTION (TYPE-1)

1 : 10

DETAIL - FLOOR TO STAIR & ELEVATOR WALL CONNECTION (TYPE-2)

1 : 10

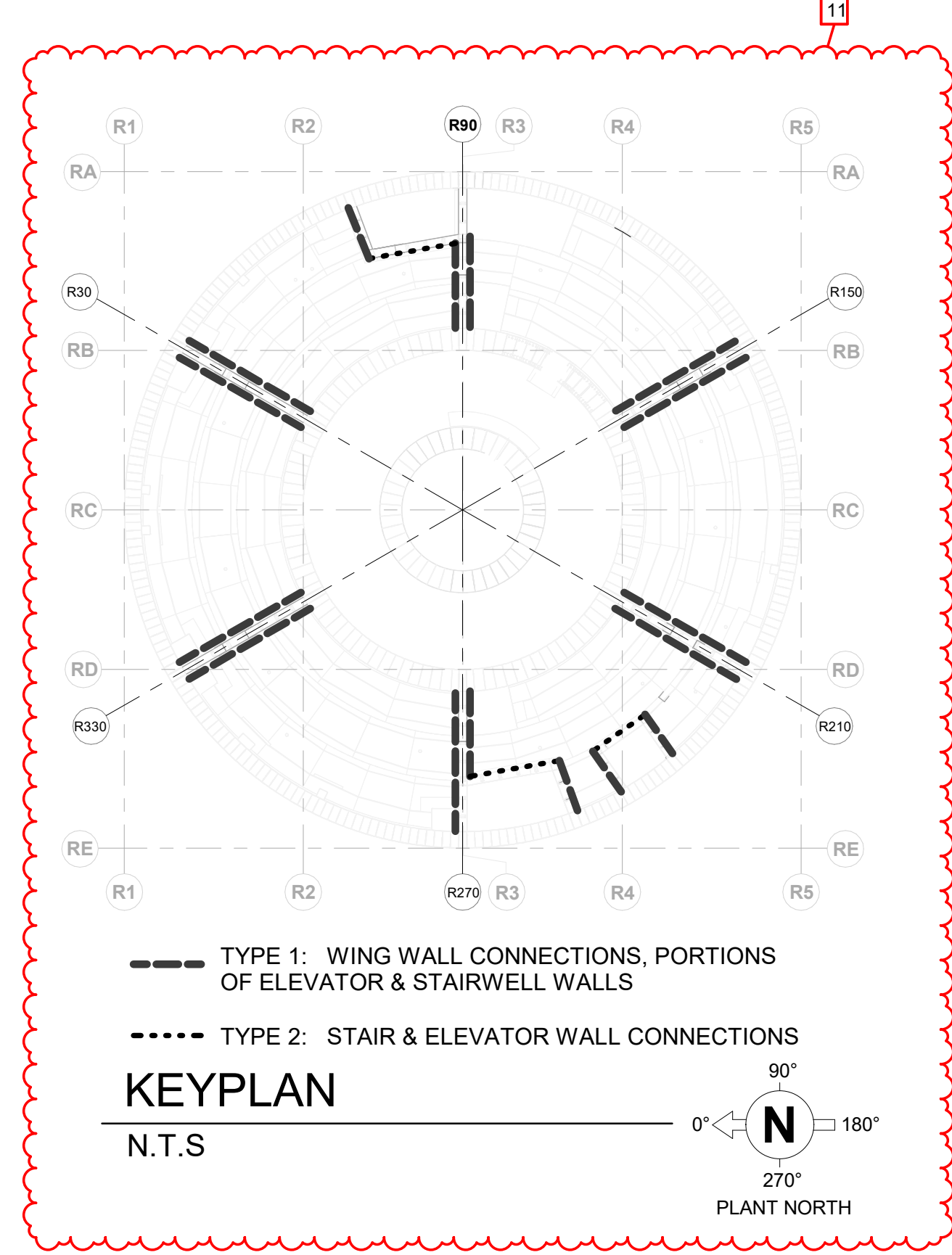


SECTION - VERTICAL STIF. PLATE AND STUDS (TYPE-1)

1 : 10

SECTION - VERTICAL STIF. PLATE AND STUDS (TYPE-2)

1 : 10



APPROVALS	DATE
DRAWN BY P. OSTROWSKI 90284142 RESPONSIBLE ENGINEER P. OSTROWSKI	10/10/2025 10/28/2025

BELOW GRADE FLOOR TO SC WALL CONNECTION DETAILS

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REVISION DATE
7/15/2026

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ISSUING AUTHORITY
CA-00060353

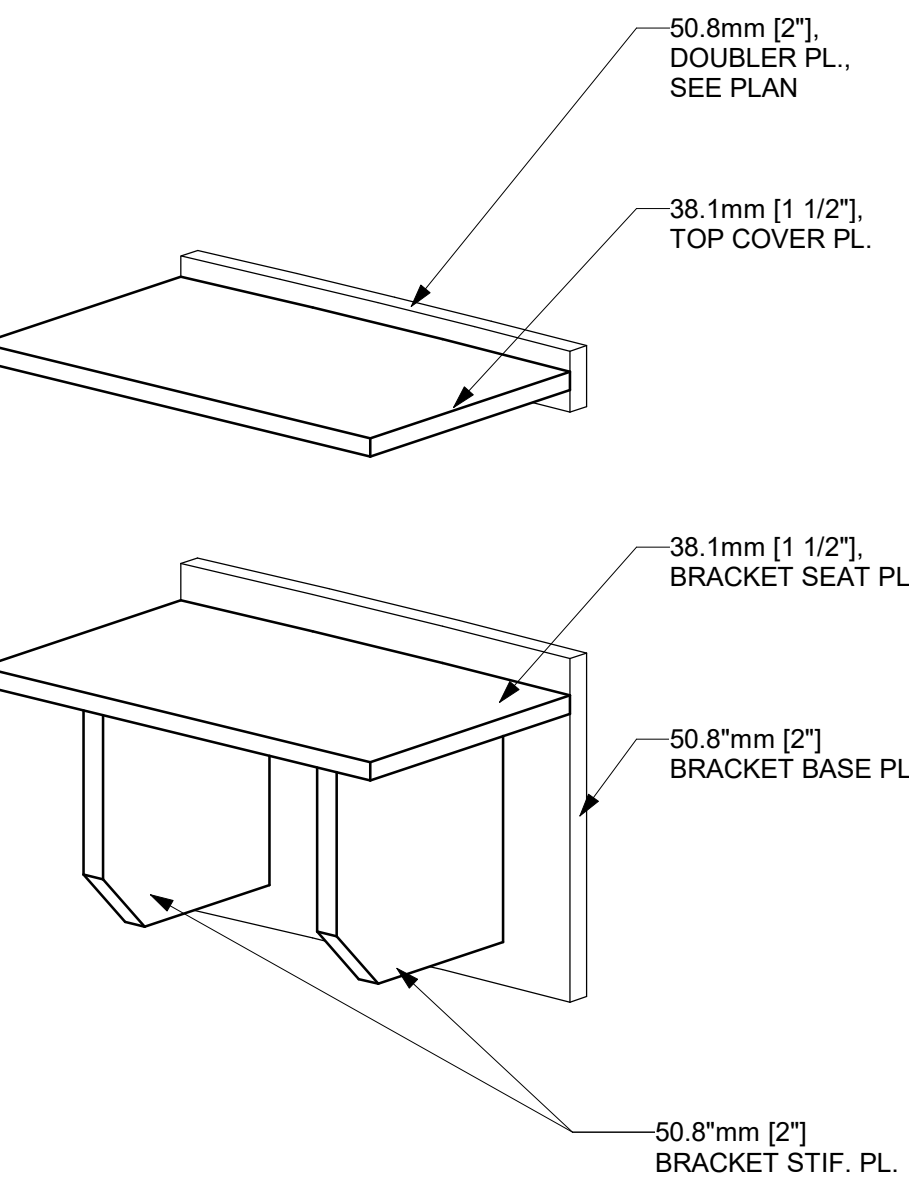
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G106

REVISIONS				
REV	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI
11	7/15/2026	R.P. LUECHT	CA-00066870 SEE SHEET G100 FOR REVISION SUMMARY	P.K.OSTROWSKI

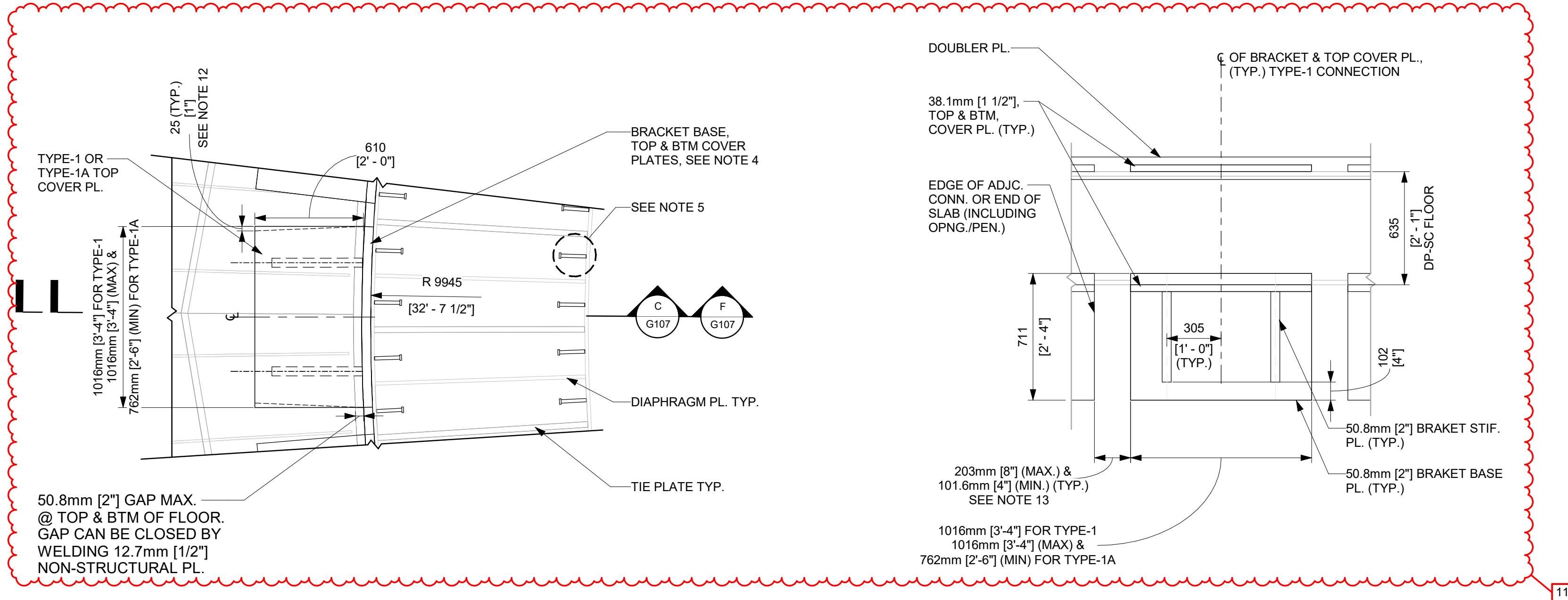
CONNECTION NOTES:

- VERIFY ALL DIMENSIONS BEFORE STARTING WORK. THE A/E AND CONTRACTING OFFICER OF DESIGNATED REPRESENTATIVE SHALL BE NOTIFIED IMMEDIATELY OF ANY DISCREPANCIES FOUND.
- WELD BEFORE POURING CONCRETE INTO FLOOR DP-SC PANEL MODULE.
- WELD CATEGORY H FOR SCCV SHAFT WALL PER TABLE 6-2 OF NEDC-33926P.
- BRACKET, BASE TOP & BLM COVER PLATES FOLLOW CURVATURE OF THE WALL.
- IF STANDARD WALL STUD ANCHORS OVERLAP THE TIE PLATES, SKIP THE STUDS THAT CONFLICT WITH THE TIE PLATES.
- TIE PLATE WELD:
 - FOR 19.1mm [3/4"] TIE PLATE:
TIE PLATE WELDS TO BE SAME AS 19.1mm [3/4"] DIAPHRAGM PLATES AT SAME ELEVATION, SEE TABLE 1-5 / G103.
 - FOR 31.8mm [1 1/4"] TIE-PLATE IN SCCV WALL, WELD SHALL BE ONE OF FOLLOWINGS:
 - (1) CJP
 - (2) DOUBLE FILLET 22.2mm (7/8")
 - (3) PJP + FILLET (EFFECTIVE THROAT SIZE OF 47.6mm [1 7/8"])
 - (4) PJP + FILLET (EFFECTIVE THROAT SIZE OF 25.4mm [1"] + SINGLE FILLET ON THE PLATE BACKSIDE (LEG SIZE OF 25.4mm [1"])CATEGORY G PER SECTION 6.13 AND FIGURE 6-3 OF NEDC-33926P.
 - FOR 31.8mm [1 1/4"] TIE-PLATE IN RB EXTERIOR WALL, WELD SHALL BE ONE OF FOLLOWINGS:
 - (1) CJP
 - (2) DOUBLE FILLET 25.4mm (1")
 - (3) PJP + FILLET (EFFECTIVE THROAT SIZE OF 50.8mm [2"])
 - (4) PJP + FILLET (EFFECTIVE THROAT SIZE OF 25.4mm [1"] + SINGLE FILLET ON THE PLATE BACKSIDE (LEG SIZE OF 25.4mm [1"])
- IN THE SCCV WALL, TIE-PLATE SHALL BE PLACED BETWEEN TWO STUD COLUMNS. IN THE RB EXTERIOR WALL, THE STUDS OVERLAPPING WITH THE TIE-PLATE SHALL BE REMOVED.

- FABRICATOR MAY ADD REINFORCING FILLETS AT THEIR DISCRETION AS LONG AS ALL OTHER DESIGN REQUIREMENTS ARE MET.
- FOR LIFTING, RIGGING, AND ALIGNMENT PURPOSES, THE CONSTRUCTOR MAY ADD ONE TAPPED HOLE TO EACH BRACKET SEAT PLATE AND ASSOCIATED TOP COVER PLATE FOR TYPE 1, TYPE 2, AND TYPE 2A BRACKETS. THE HOLE SHALL BE LOCATED AT THE CENTER OF GRAVITY OF THE PLATE, WITH A MAXIMUM DIAMETER OF 3/4 IN. AND A MAXIMUM DEPTH OF 1 IN. FOR TYPE 1 AND TYPE 2 BRACKETS, INCLUDING ASSOCIATED TOP COVER PLATES, THE HOLE MAY REMAIN IN PLACE FOLLOWING INSTALLATION. FOR TYPE 2A BRACKETS, INCLUDING ASSOCIATED TOP COVER PLATES, THE HOLE'S THREADS SHALL BE GROUND FLUSH AND THE HOLE SHALL BE PLUG WELDED AFTER ALIGNMENT AND INSTALLATION, AND THE SURFACE SHALL BE GROUND FLUSH WITH THE SEAT PLATE OR TOP COVER PLATE, AS APPLICABLE.
- CJP CRITICAL WELDED JOINT TO BE MINIMUM 50% INSPECTED BY ULTRASONIC OR RADIOGRAPHIC EXAMINATION.
- 13 mm x 102 mm RABBET MAY BE PREPARED ALONG TOP EDGE OF BRACKET BASEPLATE FOR FIT-UP OF BRACKET SEAT PLATE.
- THE SIDES OF TOP COVER PLATE AND SEAT PLATE LOCATED RB WALL SIDE CAN BE TAPERED AS SHOWN UP TO 25.4mm [1"] AS NEEDED.
- WHEN CONNECTION IS LOCATED NEXT TO SC WALL, MAXIMUM SPACING TO THE WALL CAN BE INCREASED UP TO 254mm [10"] (+/-1") TO CLEAR PATCH PLATE, UNO.
- FOR TYPE-1A, ACTUAL WIDTH SHALL BE SHOWN ON PLAN DRAWING. EXISTING TYPE-1 CONNECTIONS CAN BE MODIFIED BY TRIMMING THE SIDES OF CONNECTION (SYMETRICALLY OR UNSYMETRICALLY). THE MODIFIED DIMENSIONS ARE SHOWN ON PLAN DRAWING. IF ADDITIONAL CASE IS REQUIRED, ENGINEERING APPROVAL IS REQUIRED.
- FOR TYPE-2, IF THE CLEARANCE BETWEEN GUSSET PLATES FROM TWO ADJACENT TIE PLATES IS LESS THAN 457mm [1'4"], THE MAXIMUM SPACING MAY BE INCREASED AS NEEDED TO ACHIEVE THE REQUIRED 457mm [1'-6"] CLEARANCE BETWEEN THE GUSSET PLATES.

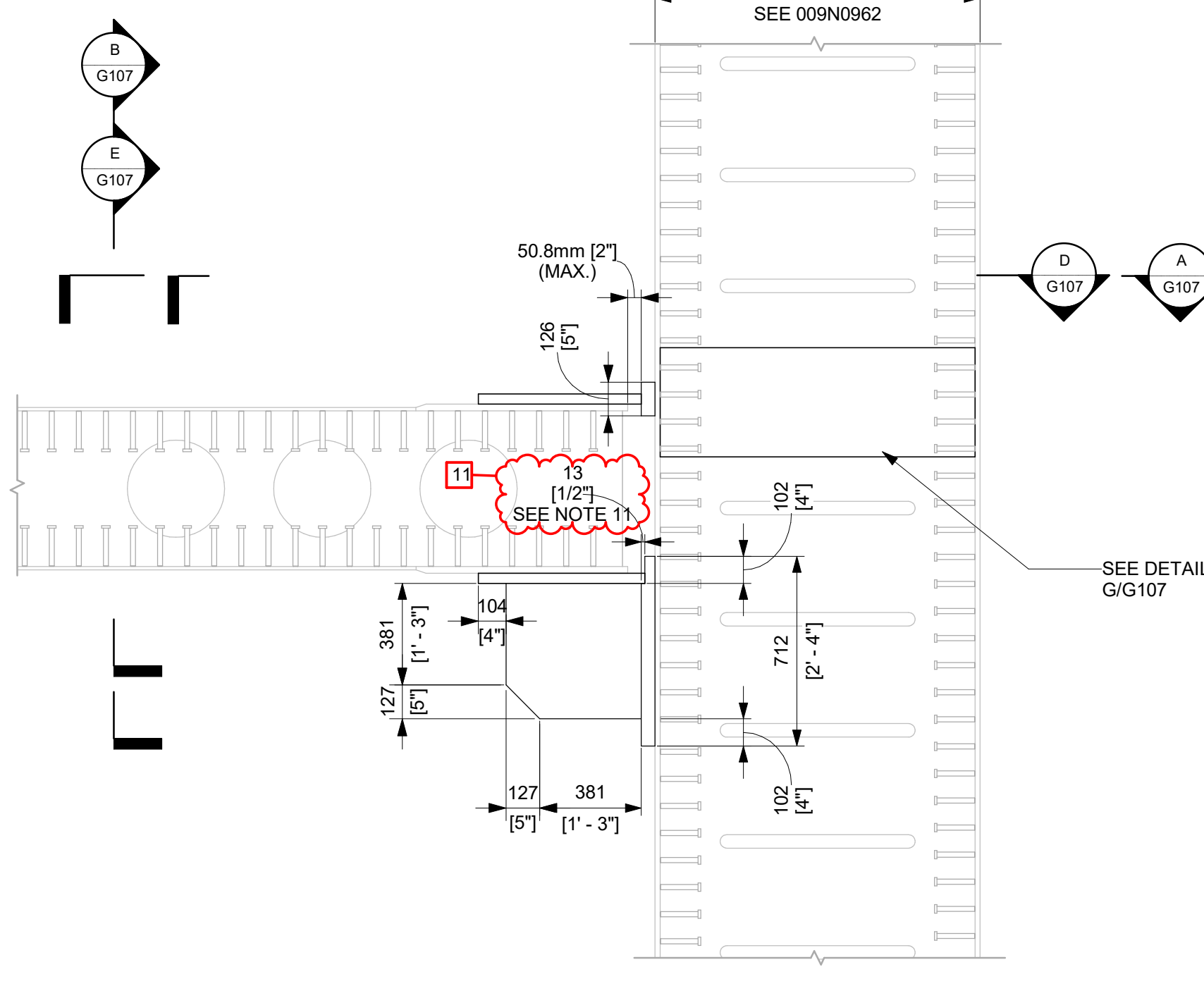


TYPE 1 & 1A:
3D VIEW

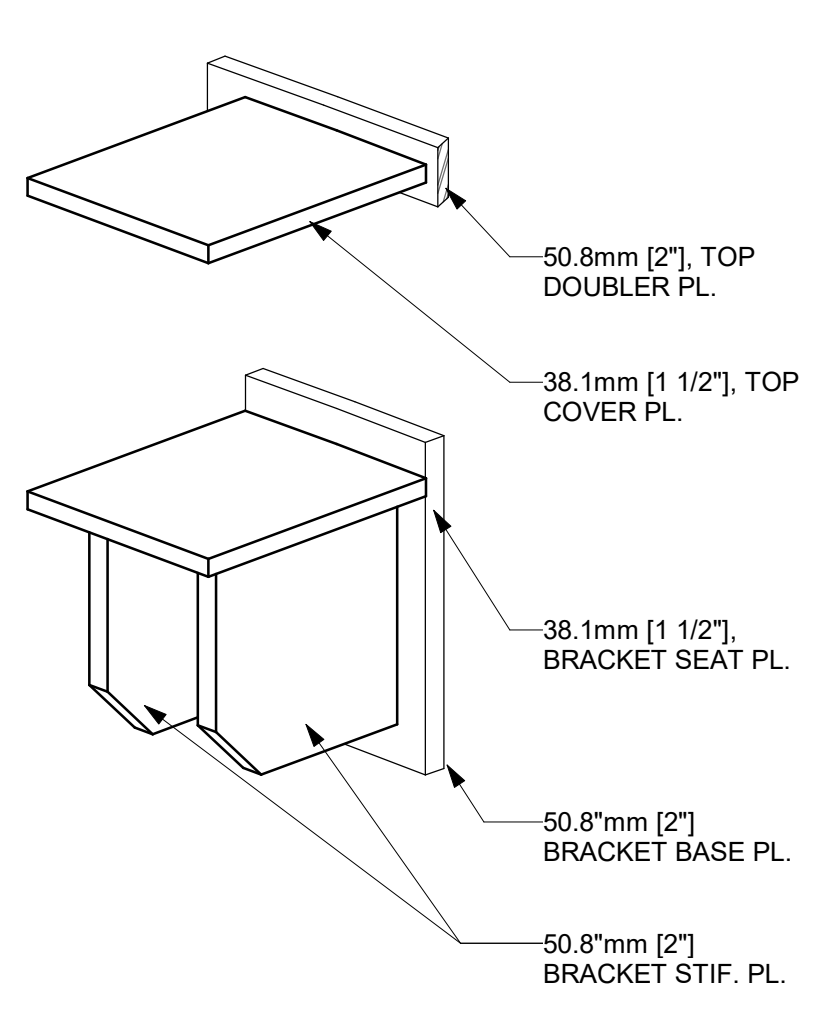


A FLOOR TO WALL CONNECTION PARTIAL PLAN (TYPE 1 & 1A)
1 : 20

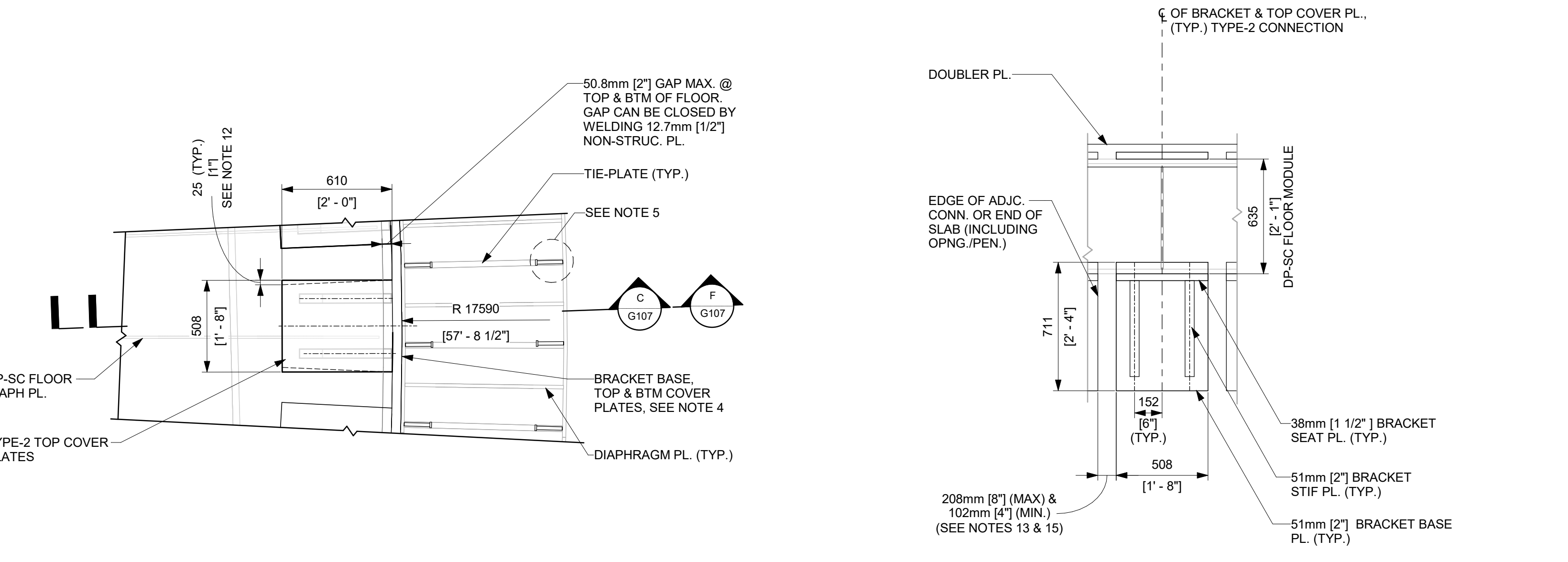
B FLOOR TO WALL CONNECTION FRONT VIEW (TYPE 1 & 1A)
1 : 20



C SECTION - FLOOR TO WALL CONNECTION - DIMENSIONS
1 : 20

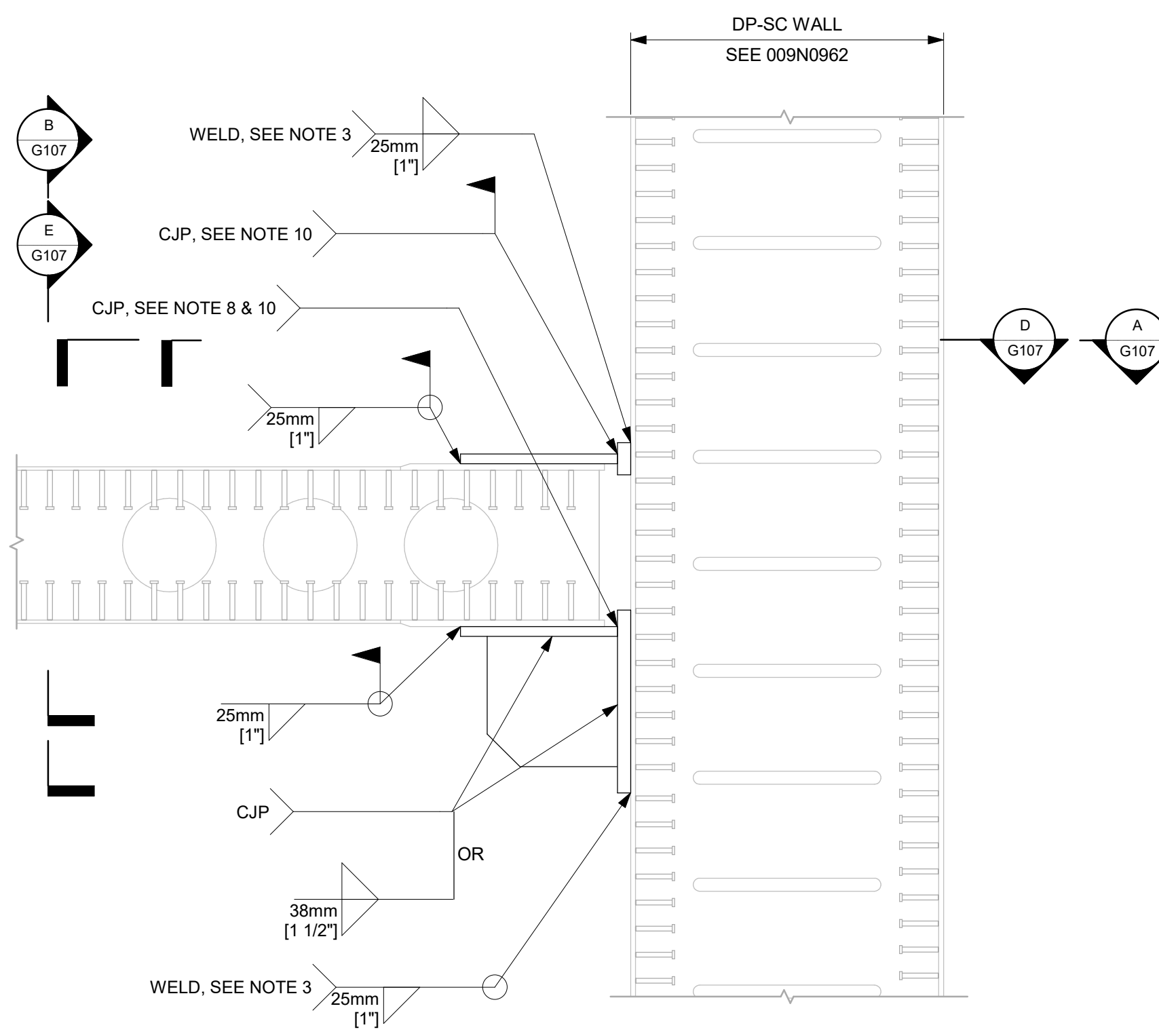


TYPE-2: 3D VIEW

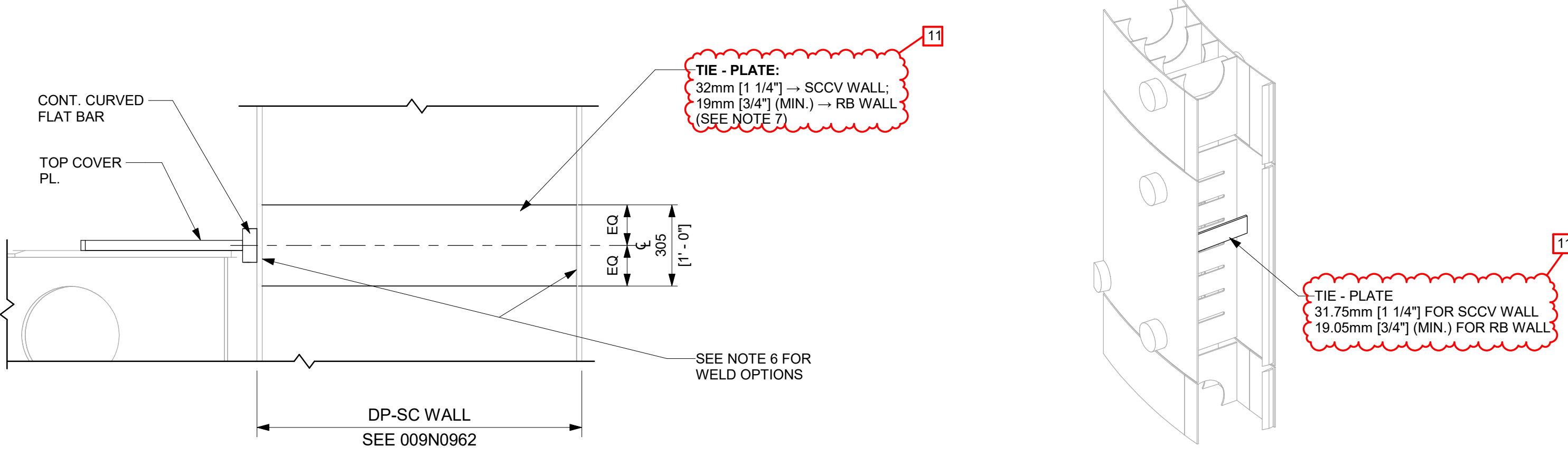


D FLOOR TO WALL CONN. PARTIAL PLAN (TYPE 2)
1 : 20

E FLOOR TO WALL CONN. FRONT VIEW (TYPE 2)
1 : 20



F SECTION - FLOOR TO WALL CONNECTION - WELD DETAIL
1 : 20



G SECTION - TIE PLATE DETAIL
1 : 15

TYP. TIE PLATE
ARRANGEMENT

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APPROVALS	DATE
DRAWN R. P. LUECHT RESPONSIBLE ENGINEER P. K. OSTROWSKI	10/10/2025 10/28/2025
SHEET TITLE BELOW GRADE FLOOR TO SHAFT WALL CONNECTION DETAILS (TYPES 1 & 2)	

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DRAWING NO. 008N9536	REVISION DATE 7/15/2026
ISSUING AUTHORITY CA-00060353	REV 11 G107



- (1) CJP
(2) DOUBLE FILLET (LEG SIZE 22.2mm [7/8"])
(3) PJP + FILLET (EFFECTIVE THROAT SIZE OF 47.6mm [1 7/8"])
(4) PJP + FILLET (EFFECTIVE THROAT SIZE OF 25.4mm [1"]) + SINGLE FILLET ON THE PLATE BACKSIDE (LEG SIZE OF 25.4mm [1"]) CATEGORY G PER SECTION 6.13 AND FIGURE 6-3 OF NEDC-33926P TYP.
6. CJP WELDED JOINT TO BE MIN 50% INSPECTED BY ULTRASONIC OR RADIOGRAPHIC EXAMINATION.



SHEET TITLE
BELOW GRADE FLOOR TO
SHAFT WALL CONNECTION
DETAILS (TYPE-3)



Headed Dowels ⁽²⁻⁴⁾										Doubler Plate ⁽¹⁾						
ELEVATION	Dowel Dia.		Dowel Length		Filllet Weld Leg Size w. dowel		Dowel Head Dia. D head min.		Headed Rebar Spacing S dowel		Thickness t doubler		Width W doubler		Filllet Weld Leg Size w. doubler	
	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm
-34.0m	1	25.4	19	482.6	0.75	19.05	3	76.2	15	381	1.25	31.75	6	152.4	0.5	12.7

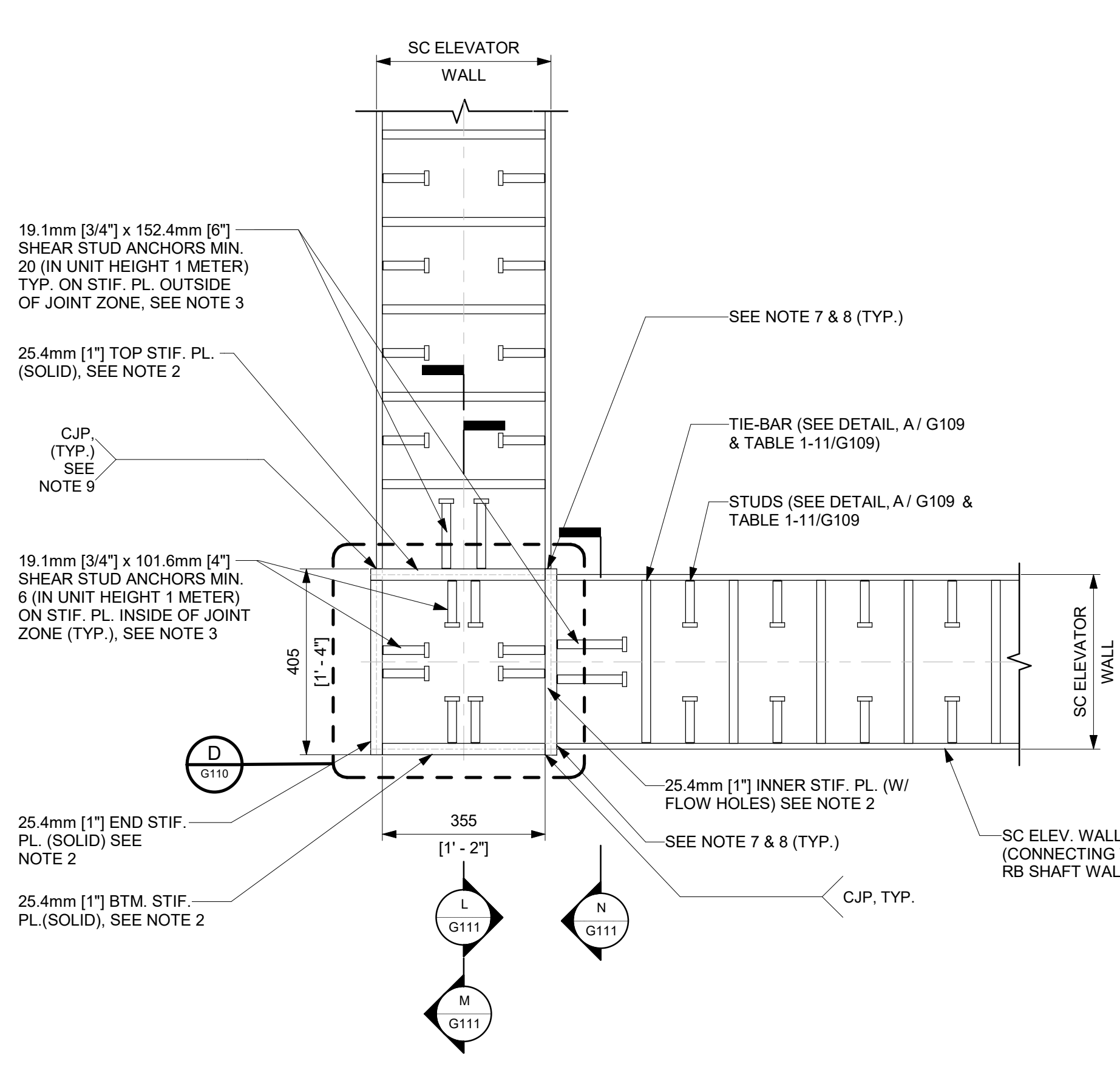
NOTES:

- ASTM 572 Grade 65 is used for the doubler plate.
- Deformed headed rebars (with threaded heads) to conform to ASTM A570, including Annex A1 requirements for Class HA head dimensions.
- Use grade 60 ksi bars for headed bars.
- Use weldable half-circle for corner application. Minimum corner outside diameter 41.275mm (1.625 in).

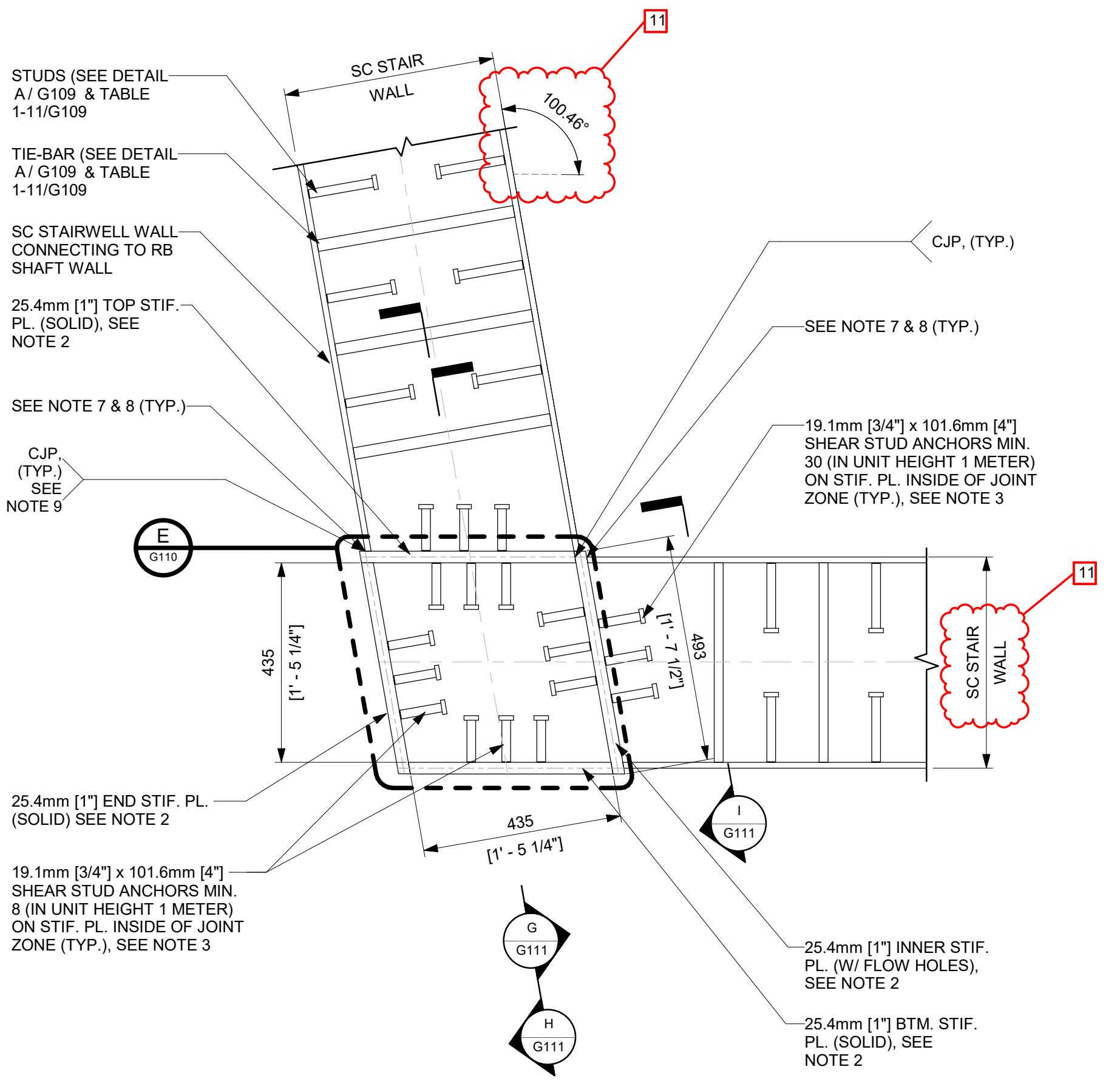
ELEVATION	Bracket ⁽²⁾										Stiffener											
	Width		Depth		Length		Thickness		Fillet Weld Leg Size		Spacing		Thickness		Depth		Fillet Weld Leg Size		Spacing		C _{se}	
	W ₁		d ₁		b ₁		t ₁		L ₁		S ₁		t _s		d _s		L _s		S _s		C _{se}	
	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm
-34.0m	6	152.4	6	152.4	6	152.4	0.5	12.7	0.375	9.5	15	381	0.5	12.7	5.5	139.7	0.375	9.5	4.75	120.7	1	25.4
NOTES:																						
1) Top bracket material shall be ASTM 572 Grade 50 for all top connection components.																						
2) Angles ASTM 572 Grade 50 can be used for top bracket.																						

REVISIONS				
REV	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI
11	7/15/2026	R.P. LUECHT	CA-00066870 SEE SHEET G100 FOR REVISION SUMMARY	P.K.OSTROWSKI

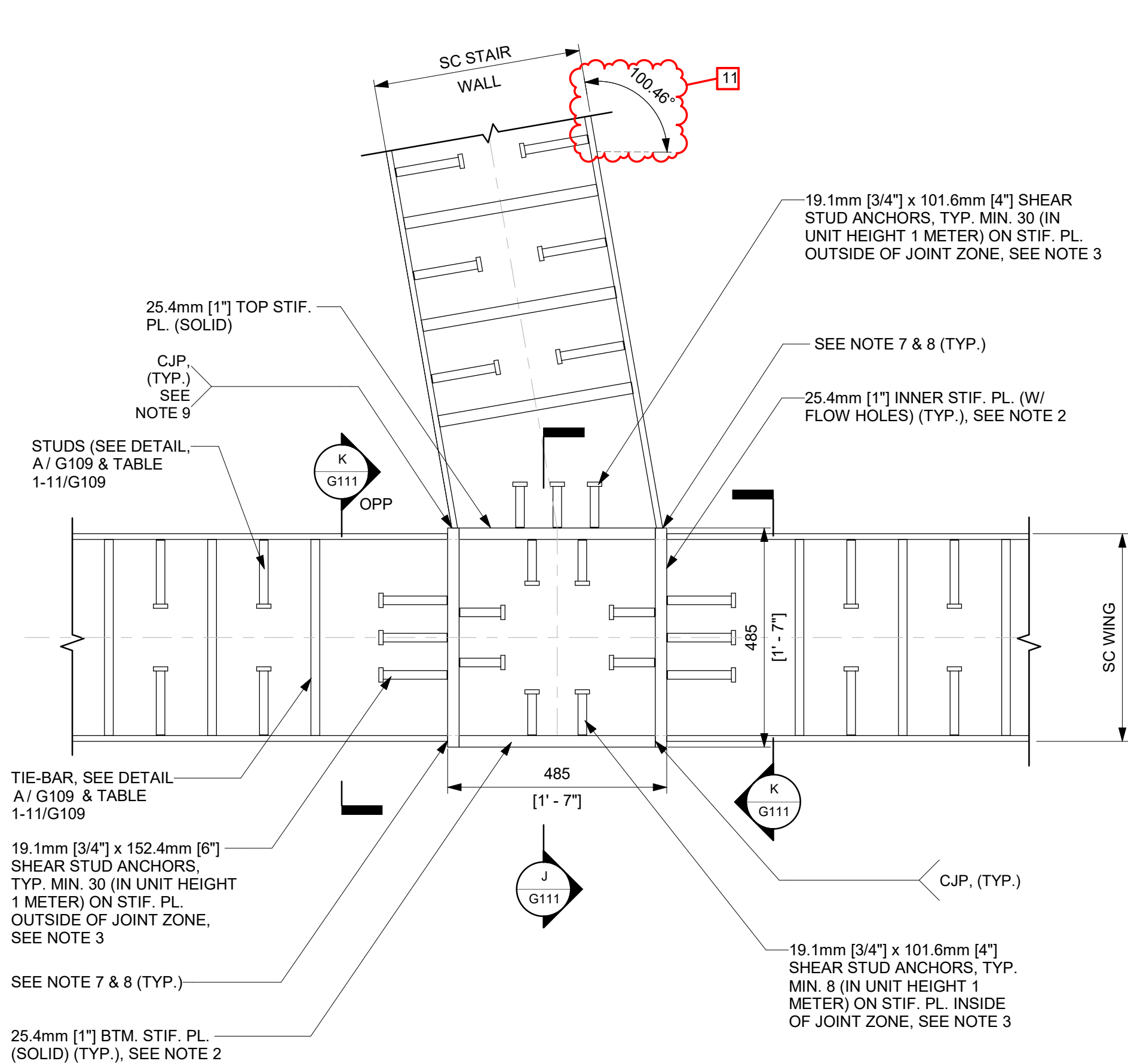
- NOTES:**
1. VERIFY ALL DIMENSIONS BEFORE STARTING WORK. THE A/E CONTRACTING OFFICER OF DESIGNATED REPRESENTATIVE SHALL BE NOTIFIED IMMEDIATELY OF ANY DISCREPANCIES FOUND.
 2. STIFFENER PLATE SHALL BE ASTM A572 GRADE 65.
 3. STUDS MINIMUM SPACING IS 76.2mm [3"]. SHIFT STUDS AS NEEDED TO AVOID INTERFERENCE. MAINTAIN MINMAX SPACING. ADJUST STUD QUANTITY PROPORTIONALLY TO ACTUAL SPACING.
 4. IF A VERTICAL WALL SPLICE EXISTS BETWEEN TWO CONCRETE FLOW HOLES AND THE SPACING BETWEEN THE HOLES DEVIATES FROM THE SPECIFIED UNIT LENGTH (1 METER), ADJUST THE NUMBER OF SHEAR STUDS BETWEEN THE CONCRETE FLOW HOLES PROPORTIONALLY, BASED ON THE RATIO OF THE ACTUAL SPACING TO THE SPECIFIED UNIT LENGTH.
 5. SEE SHEETS S105, S401, S402, & S403 FOR REFERENCE.
 6. SEE 008N9536, SHEET G109 FOR GENERAL DETAILS OF SC WALLS.
 7. STAIR, WING AND ELEVATOR WALL FACEPLATES SHALL BE ALIGNED AND FALL WITHIN THE CROSS-SECTIONAL AREA OF THE CORRESPONDING STIFFENERS, AS SHOWN IN THE DRAWING DETAIL.
 8. THE ENDS OF THE SC WALL-TO-WALL CONNECTING FACEPLATES (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) SHALL BE TRIMMED AND DIRECTLY ATTACHED TO THE NEXT ADJACENT WALL MODULE. AT THE CONSTRUCTOR'S DISCRETION, A PATCH PLATE CAN BE USED TO EXTEND THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) WHEN A GAP IS PRESENT ON SITE. THE PATCH PLATE'S THICKNESS AND MATERIAL SHALL BE IDENTICAL TO THE SC WALL'S FACEPLATE (OR 25.4mm [1"] VERTICAL STIFFENER PLATE) AND SHALL BE CONNECTED TO THE FACEPLATE USING A CJP WELD.
 9. FIELD WELDING MAY BE PERFORMED AS REQUIRED BY THE FABRICATION SEQUENCE.



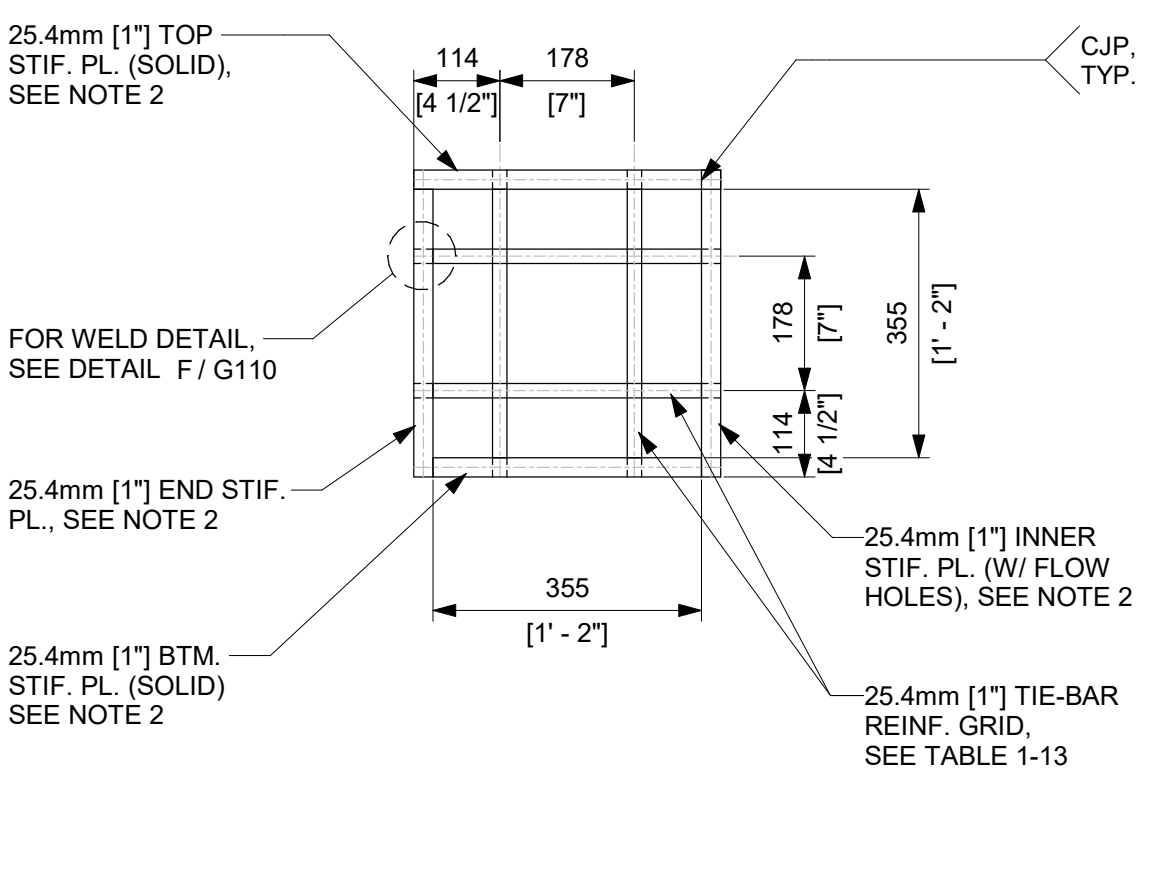
A SC ELEVATOR WALL-TO-WALL CONNECTIONS
1 : 10 SEE SHEET S403



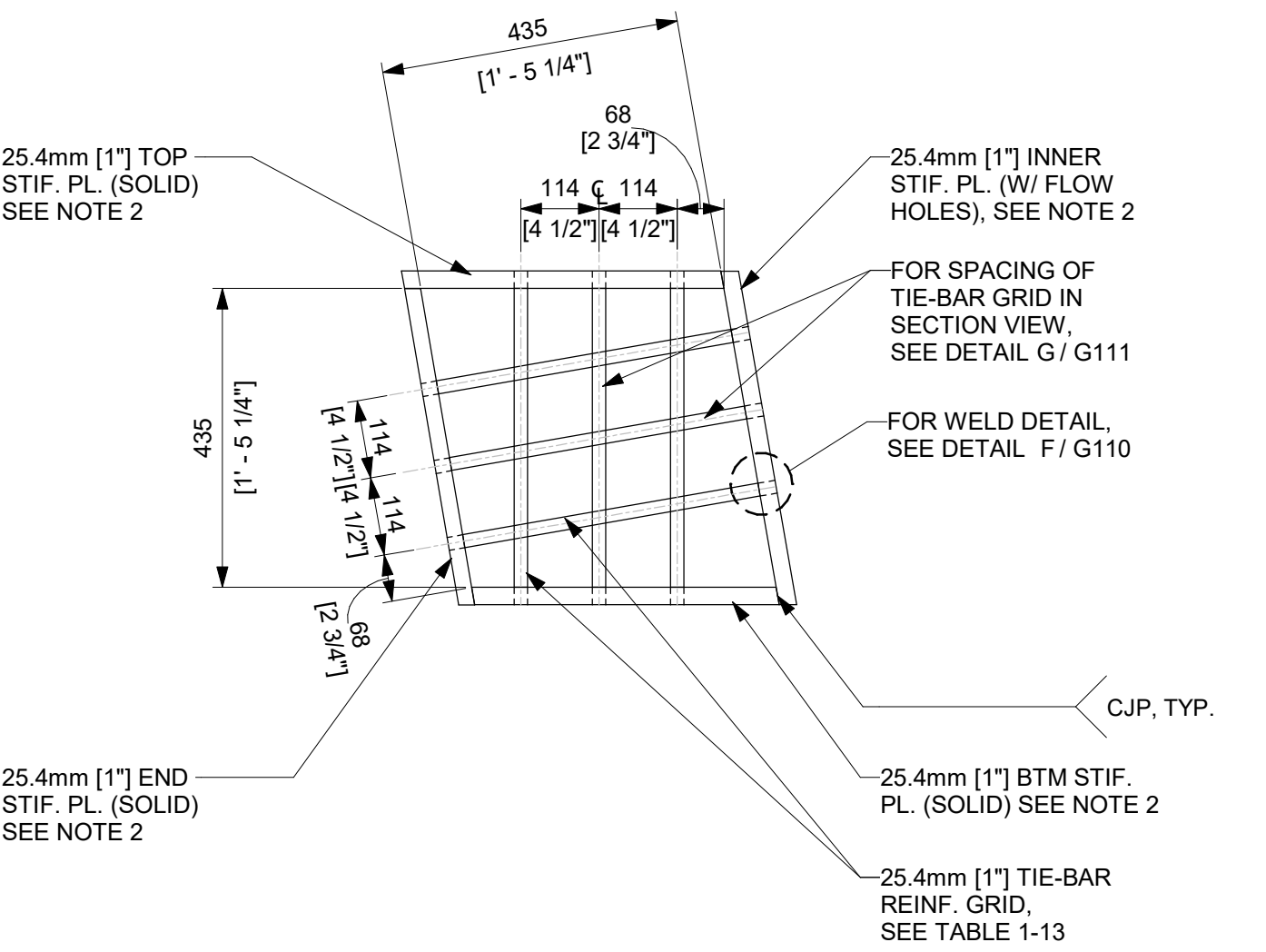
B SC STAIRWELL WALL-TO-WALL CONNECTIONS @ STAIR A AND B LOCATIONS
1 : 10 SEE SHEETS S401 & S402



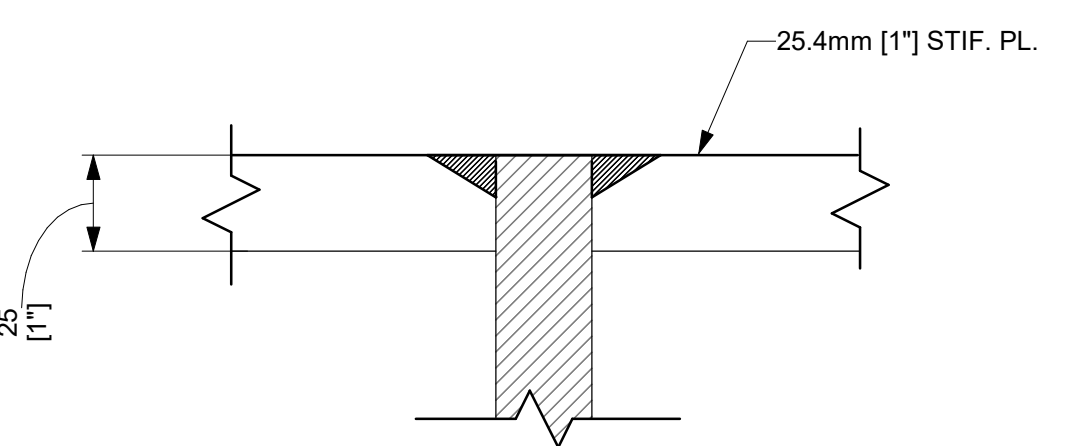
C SC STAIRWELL WALL-TO-WING CONNECTIONS @ STAIR A AND STAIR B LOCATIONS
1 : 10 SEE SHEETS S401 & S402



D ENLARGE - ELEVATOR TIE-ROD DETAIL
1 : 10 SEE SHEETS S401 & S402



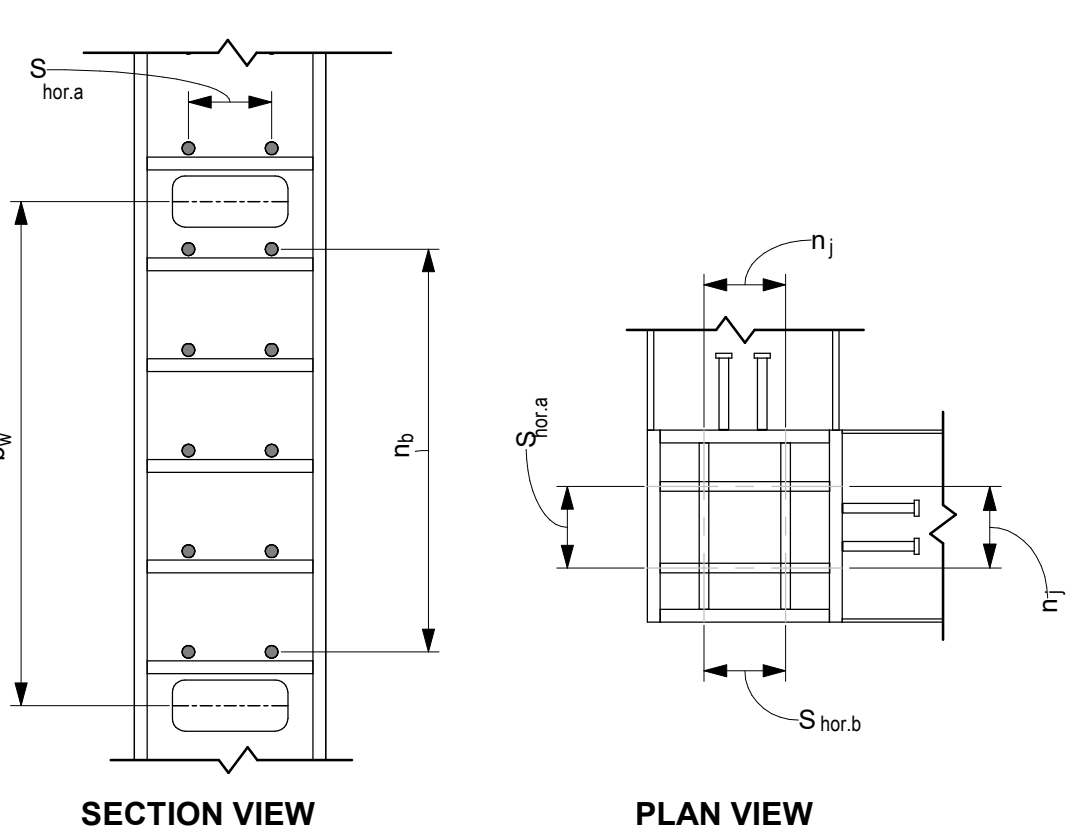
E ENLARGE - STAIRWELL TIE-ROD DETAIL
1 : 10 SEE SHEETS S401 & S402



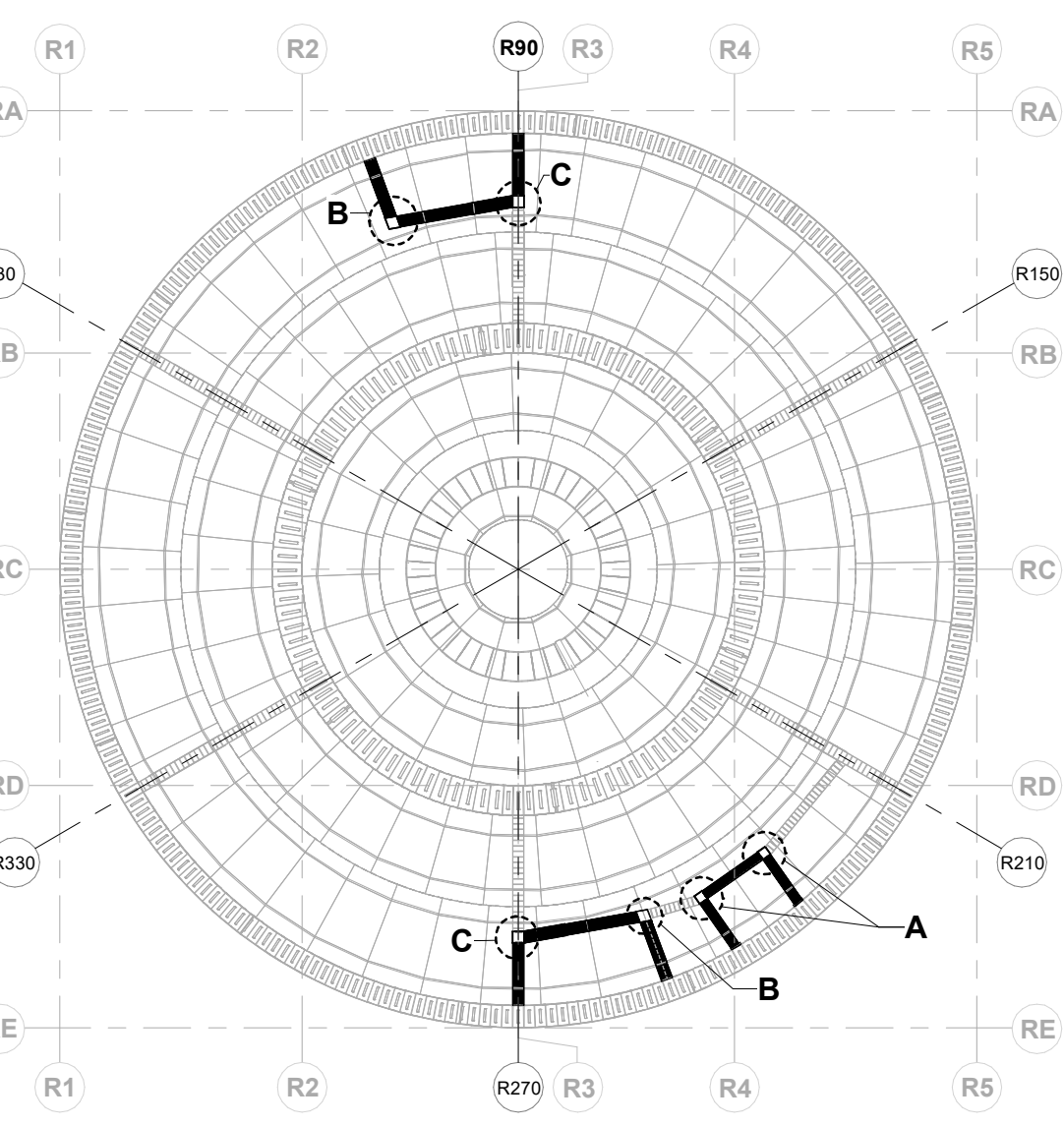
F PJP WELD DETAIL
1 : 2
NOTES:
1. SEE TABLE 1-13 FOR WELD SIZE
2. TIE-BARS TO BE FLUSH W/ STIFFENER PLATE

TABLE 1-14														
SC WALL	Tie-Bar Diameter	Spacing In Direction of Joint Depth		Spacing In Direction of Unit Height		Number of Tie-Bars In the Direction of Joint Depth	Number of Tie-Bars In Direction of the Unit Height (equal to number of tie-bar layers)	Unit Height (c)		PJP Weld Effective Throat Size	Tie-Bar Yield Strength, F_y	Tie-Bar Tensile Strength, F_t		
		$S_{hor,a}/S_{hor,b}$		S_v				b_w					S_w	
		(mm.)	(in.)	(mm.)	(in.)			(mm.)	(in.)	(mm.)	(in.)	(mm.)	(in.)	
Elevator Wall	25.4	1.0	177.8/177.8	7.0/7.0	200.0	7.9	2 ^(b)	5 ^(c)	1000	39.4	9.525	0.375	50 (345)	65 (448)
Stairwell Wall A&B	25.4	1.0	114.3/114.3	4.5/4.5	250.0	13.1	3 ^(b)	4 ^(c)	1000	39.4	9.525	0.375	50 (345)	65 (448)

Note:
a) Tie-bars are ASTM A572 Grade 50.
b) n_j represents the number of tie-rods in both directions forming a 2X2 and 3X3 grid for the elevator and stairwell wall-to-wall connections respectively.
c) n_b represents the number of tie-rod grid layers per meter (unit height)



TIE-BAR REINFORCEMENT DIMENSIONS
N.T.S



KEYPLAN - CONNECTIONS
N.T.S
90°
0°
180°
270°
PLANT NORTH

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ENGR: PETER OSTROWSKI
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EXPIRES: 01/31/2027

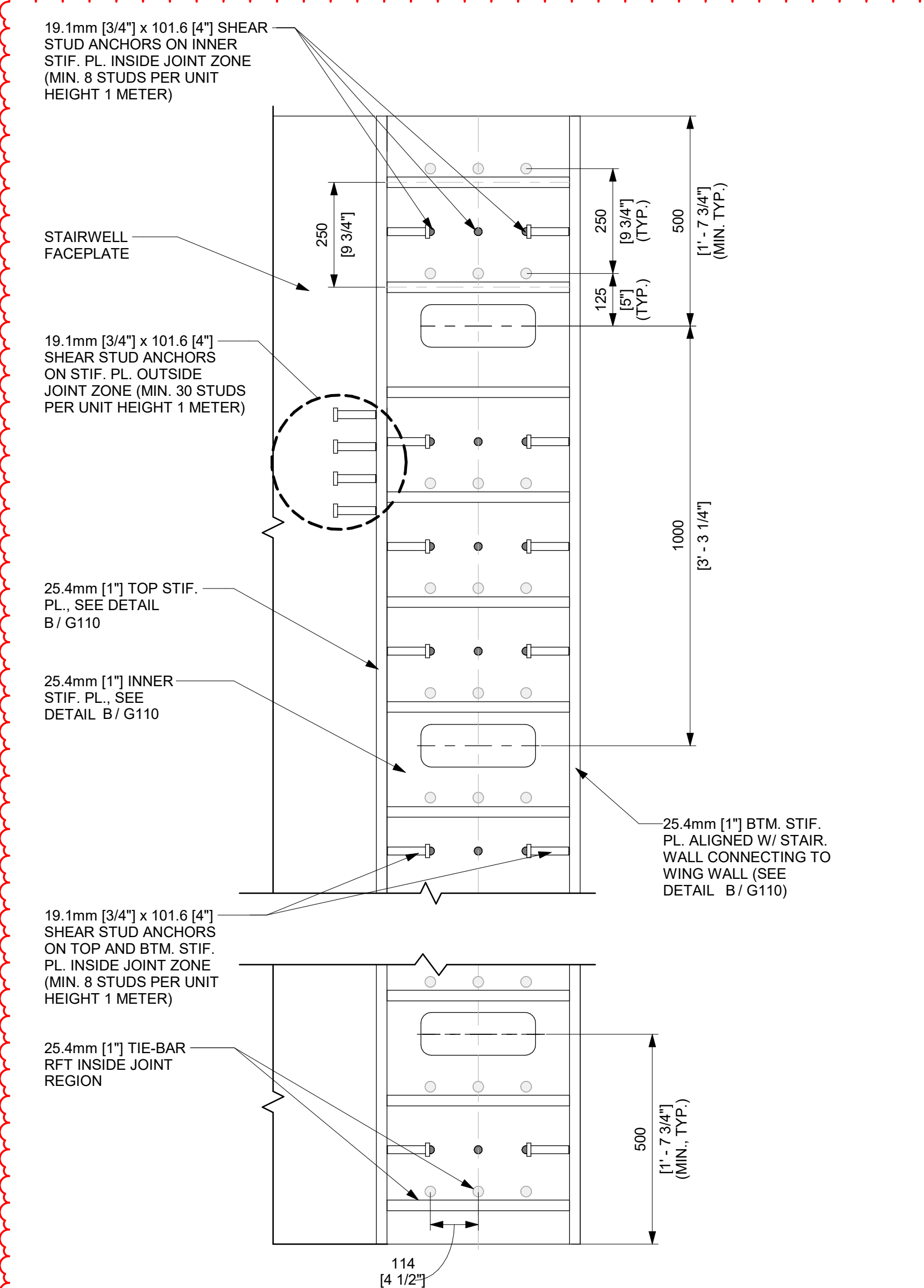
APPROVALS	DATE
R. P. LUECHT	01/25/2025
P. K. OSTROWSKI	04/15/2025

STANDARD DETAILS

DRAWING NO: **008N9536**
ISSUING AUTHORITY: CA-00061148

REVISION DATE: 7/15/2026
SHEET NO: 11
G110

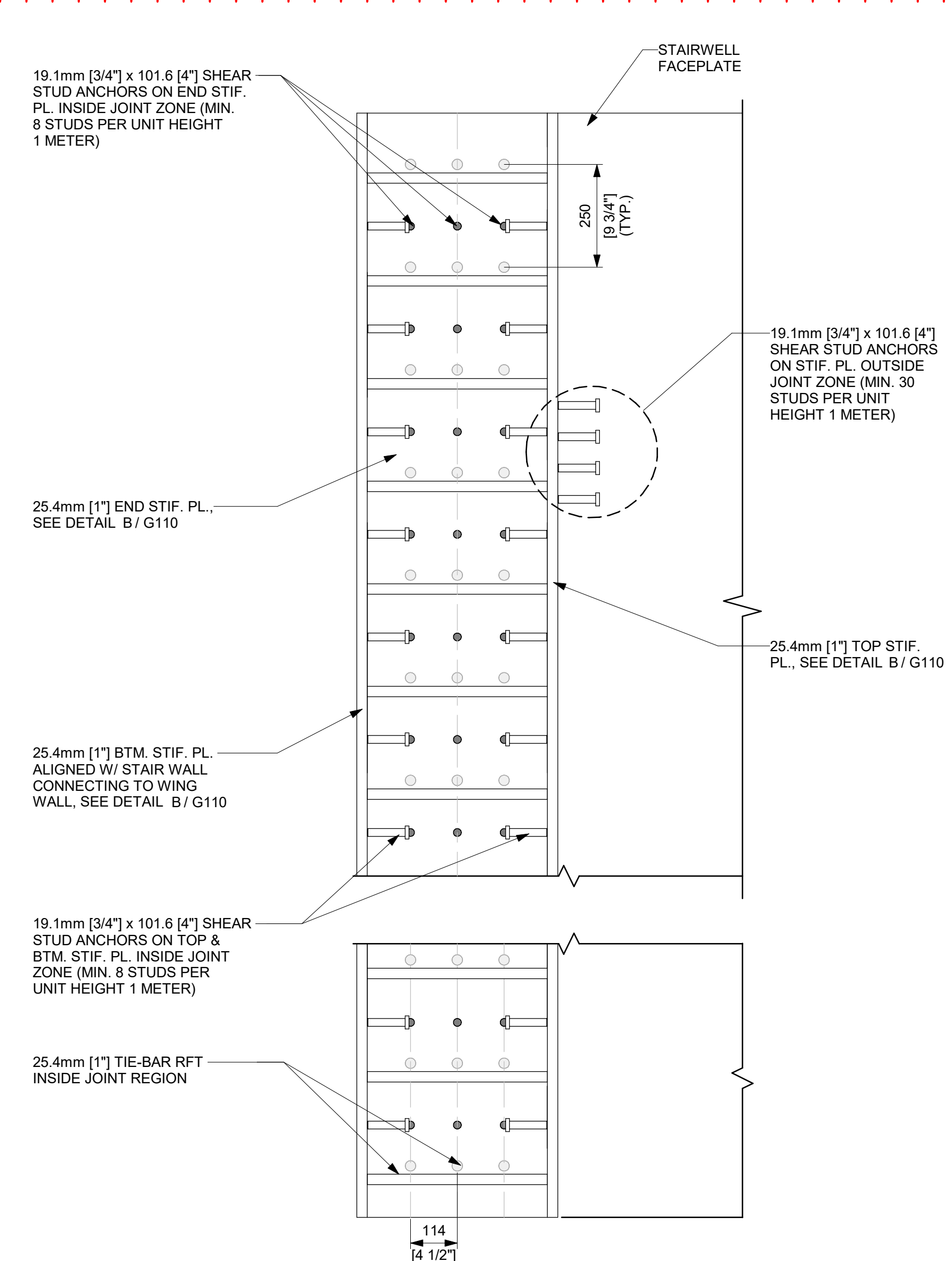
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REV/	DATE	DRAWN BY	CA NUMBER / DESCRIPTION	RESPONSIBLE ENGINEER
9	11/21/2025	R LUECHT	CA-00061148 SEE SHEET G100 FOR REVISION SUMMARY	P OSTROWSKI
10	04/06/2026	R. P. LUECHT	CA-00063942 SEE SHEET G100 FOR REVISION SUMMARY	P.K. OSTROWSKI



SECTION - STAIRWELL WALL INNER STIF. PL. STUDS AND TIE-BAR RFT. (INSIDE OF JOINT)

1 : 10

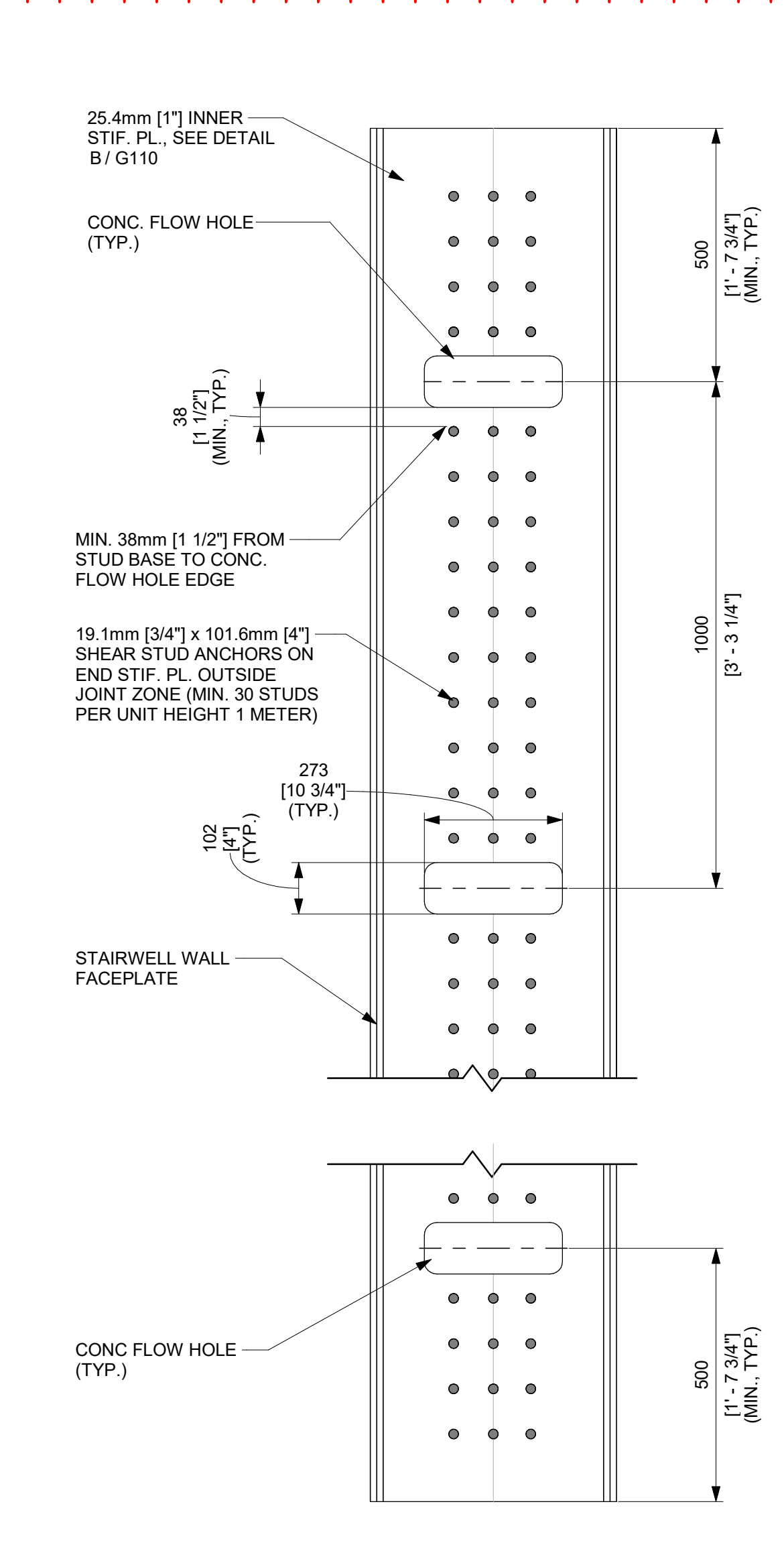
NOTE: SECTION SHOWN FLAT FOR CLARITY. NOT A TRUE PROJECTION. REFER TO DETAIL B/G110 FOR TRUE ORIENTATION.



SECTION - STAIRWELL WALL END STIF. PL. STUDS AND TIE-BAR RFT. (INSIDE OF JOINT)

1 : 10

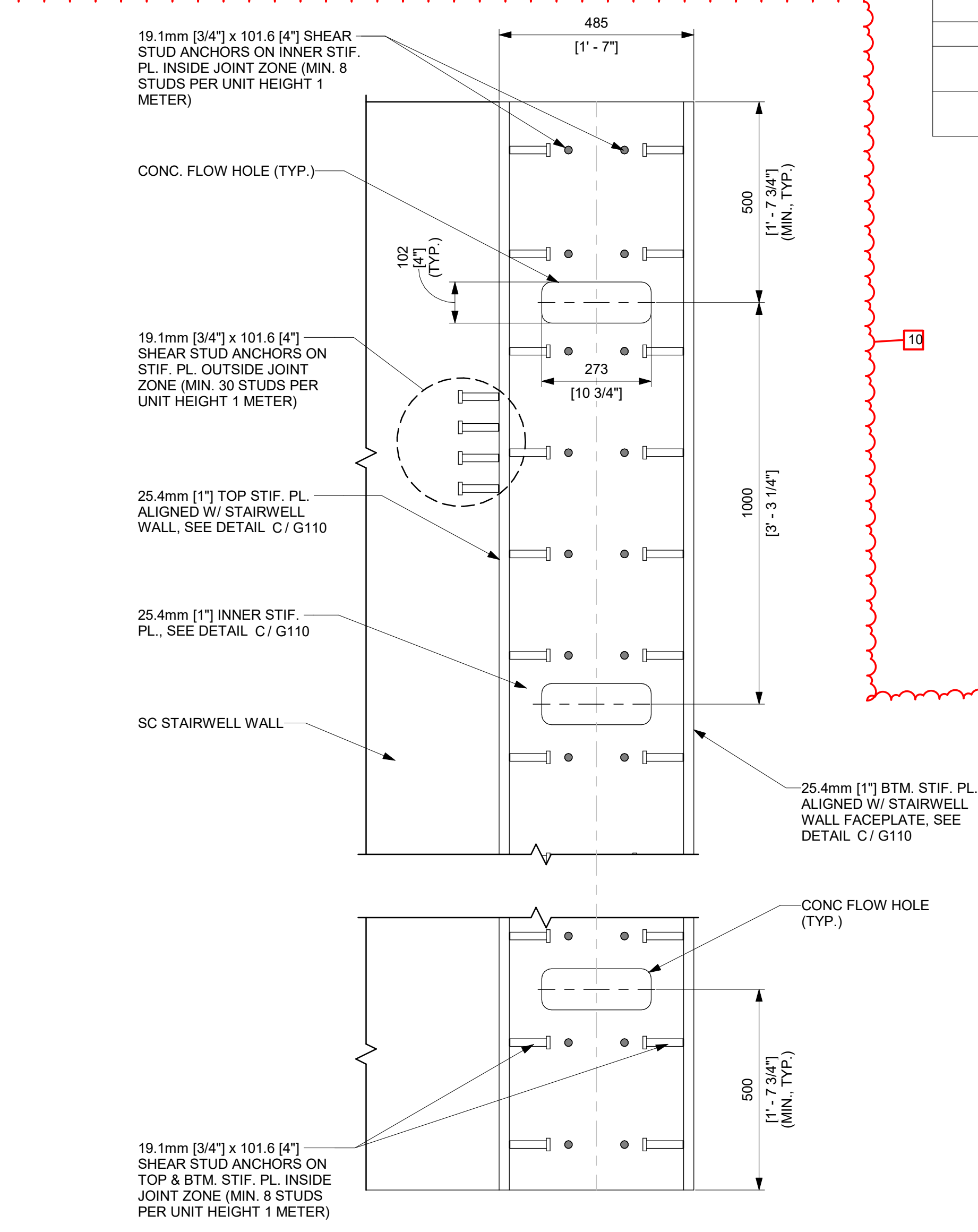
NOTE: SECTION SHOWN FLAT FOR CLARITY. NOT A TRUE PROJECTION. REFER TO DETAIL B/G110 FOR TRUE ORIENTATION.



SECTION - STAIRWELL WALL INNER STIFFENER PLATE AND STUDS (OUTSIDE OF JOINT)

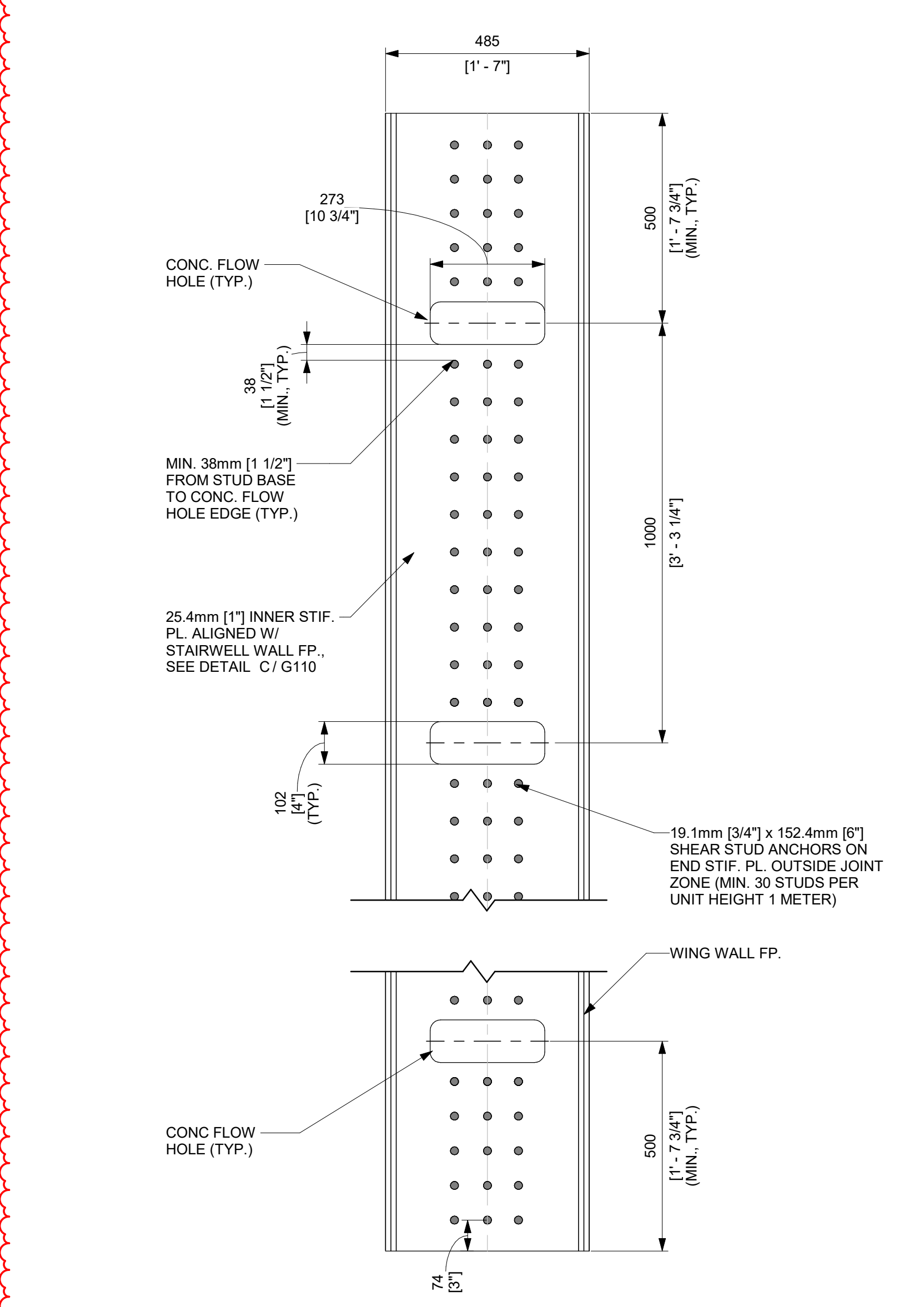
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NOTE: SECTION SHOWN FLAT FOR CLARITY. NOT A TRUE PROJECTION. REFER TO DETAIL B/G110 FOR TRUE ORIENTATION.



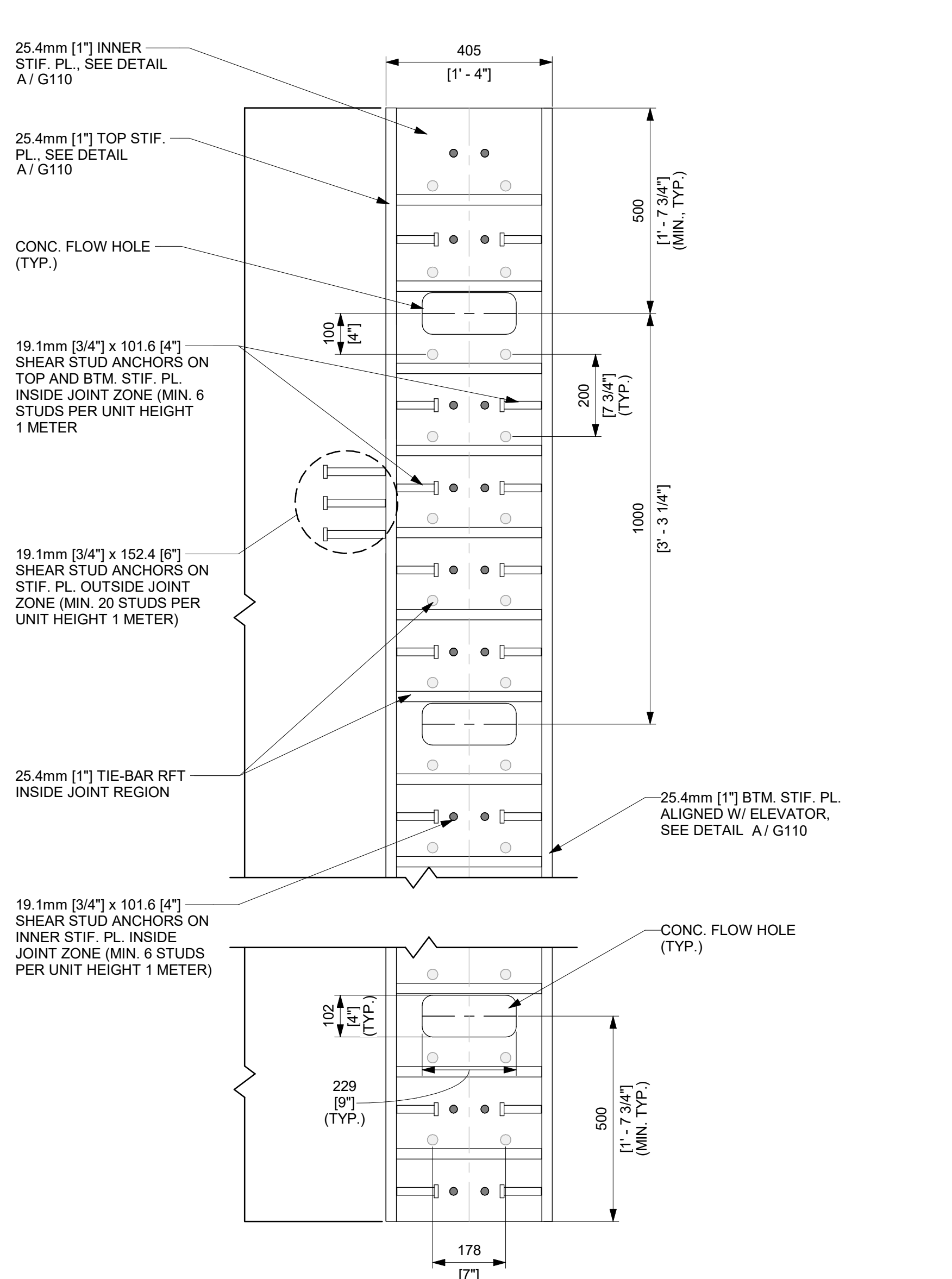
SECTION - WING WALL INNER STIF. PLATE & STUDS (INSIDE OF JOINT)

1 : 10



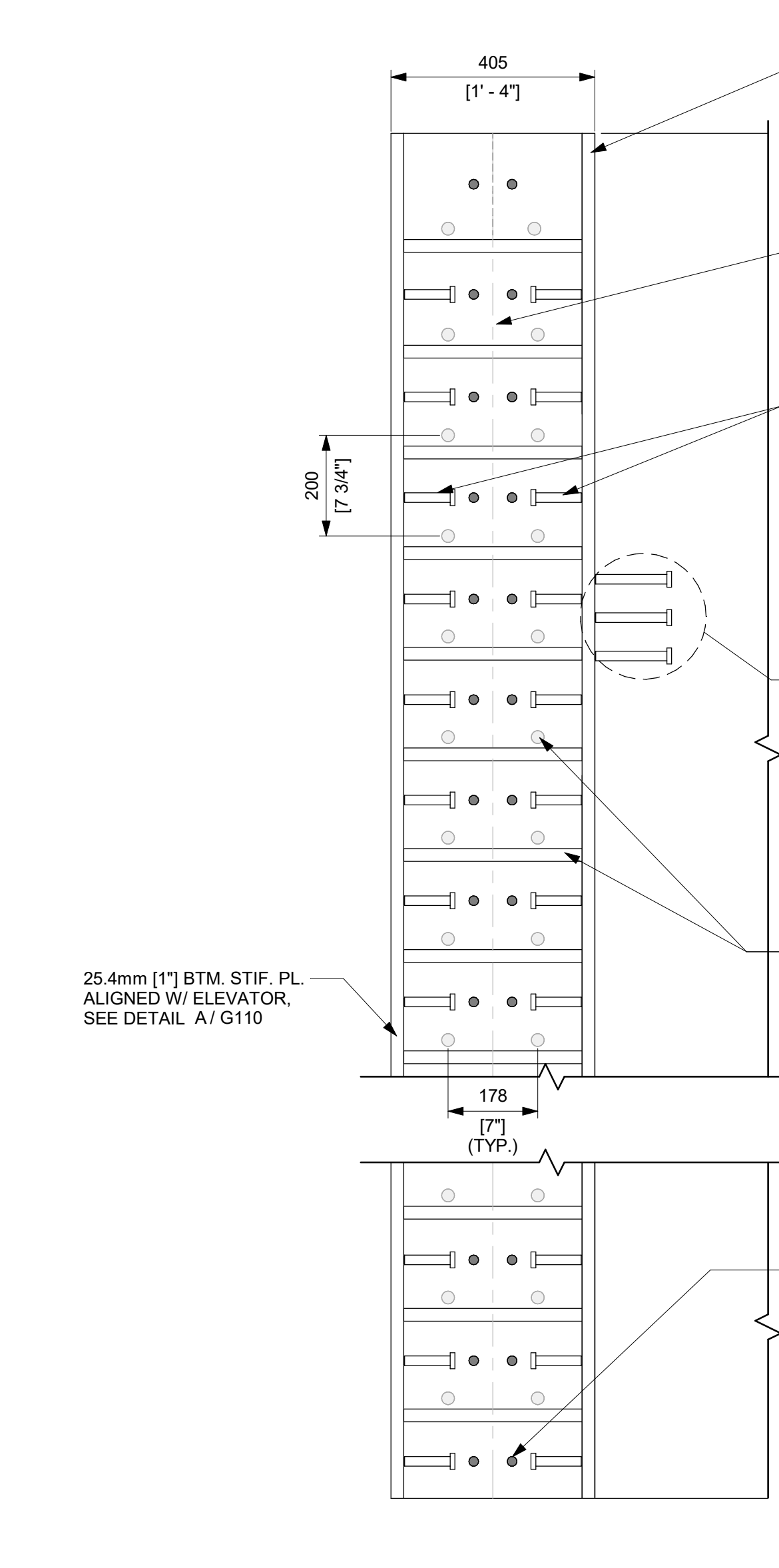
SECTION - WING WALL INNER STIF. PLATE AND STUDS (OUTSIDE OF JOINT)

1 : 10



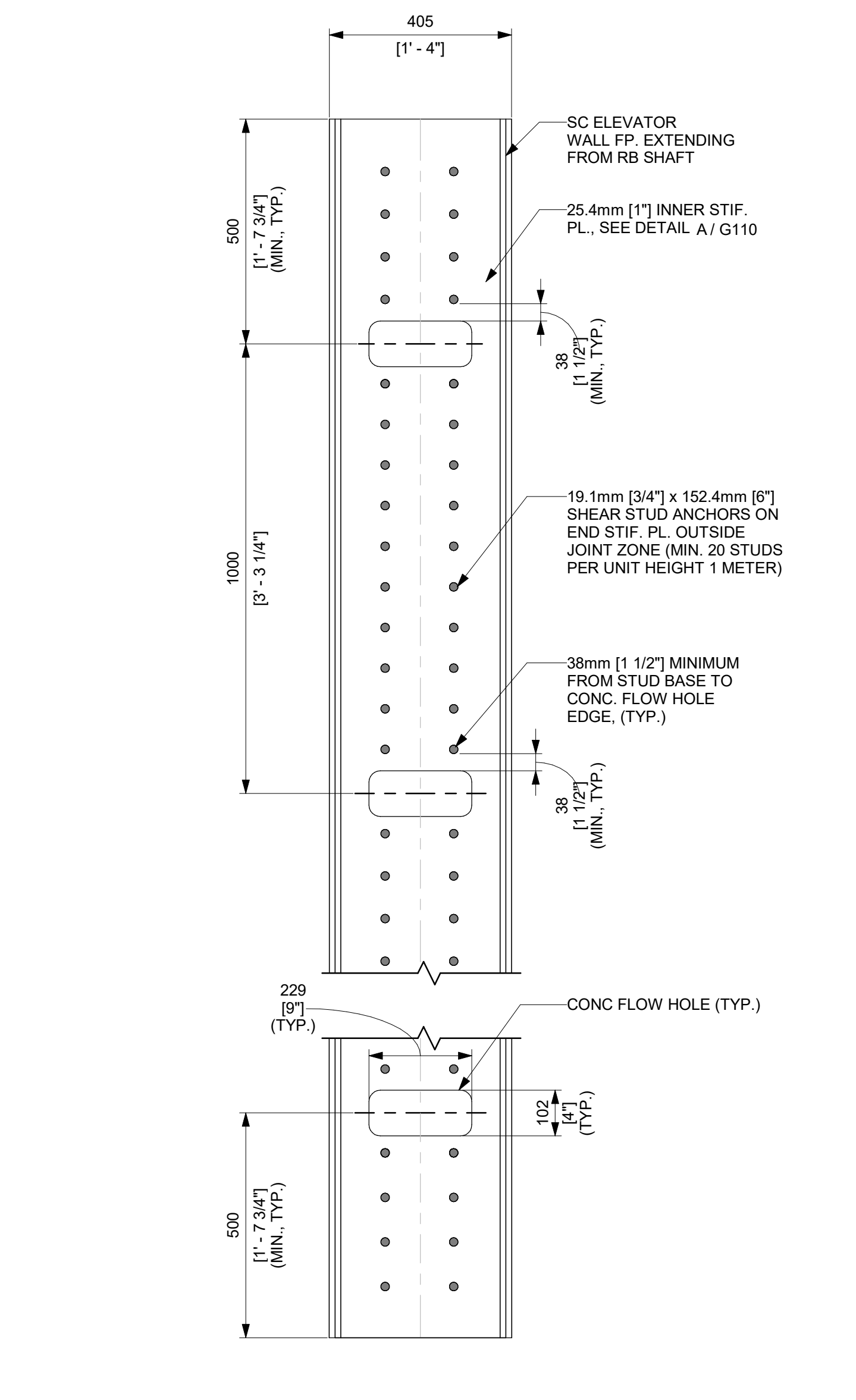
SECTION - ELEVATOR WALL INNER STIF. PL. STUDS AND TIE-BAR RFT. (INSIDE OF JOINT)

1 : 10



SECTION - ELEVATOR WALL END STIF. PLATE STUDS AND TIE-BAR RFT. (INSIDE OF JOINT)

1 : 10



SECTION - ELEVATOR WALL INNER STIF. PLATE AND STUDS (OUTSIDE OF JOINT)

1 : 10



APPROVALS	DATE
DRAWN R. P. LUECHT	01/25/2025
RESPONSIBLE ENGINEER P. K. OSTROWSKI	04/15/2025

ISSUED FOR CONSTRUCTION WITH HOLDS
04/06/2026

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